
Clinical- and Commercial-Scale Reactors

This chapter discusses the challenges in the scale-up of perfusion bioreactors from the few litres laboratory scale to the thousands of litres clinical and commercial scale. We consider comparative studies between laboratory- and large-scale reactor systems, including multiphase reactor models and computational fluiddynamic tools as well as omics studies to support solid and reliable scale-up procedures. Specific scale-up issues – such as the scalability of the cell retention device long-term operation and batch definition in the case of process failure – are discussed. Finally, we evaluate the potential of single-use technologies and close the chapter with economic, financial and environmental considerations in the context of future developments in biomanufacturing.

5.1 SCALE-UP CHALLENGES AND FUTURE PERSPECTIVES

We consider here the largest scales in perfusion bioreactors, which refer to clinical and commercial production. The objective is to implement the perfusion process at such scales while retaining the same performances obtained at the litre scale that is at the end of the process development phase discussed in Chapter 4. This means the same productivity and yield, as well as the same product quality specifications. The scale-up from the small to the larger scale is a classical chemical engineering problem, as schematically illustrated in Figure 5.1. This involves issues about gas–liquid–solid mass transport, mixing and hydrodynamic stresses, which are treated in classical books about the theory of multiphase reactors (Assirelli et al. 2005, Kawase et al. 1992, Nienow 1998, Shah 1979).

The first step is indeed to understand the fluid dynamics and demonstrate comparable behavior at different scales, which is also affected by the operating mode. While the state-of-the-art fed-batch technology uses production vessels in the range of 10 to 20×10^3 L, the perfusion technology requires, in fact, comparably smaller production volumes, which, due to the higher volumetric productivity, are in the range of 0.5 to 2×10^3 L (Bielser et al. 2018, Fisher et al. 2018, Konstantinov & Cooney 2015). On the other hand, the ancillary equipment in perfusion – in particular, the cell retention device – has to be taken into account and adds complexity to the system. For both operating modes, the litre scale bioreactor is typically considered sufficient for a safe scale-up process of the stirred vessel at commercial scale. The microenvironment in large vessels is obviously much more heterogeneous in the sense that it can change substantially in space due to local differences in mixing intensity – possibly resulting in pH, oxygen concentration and temperature differences – and consequently also in

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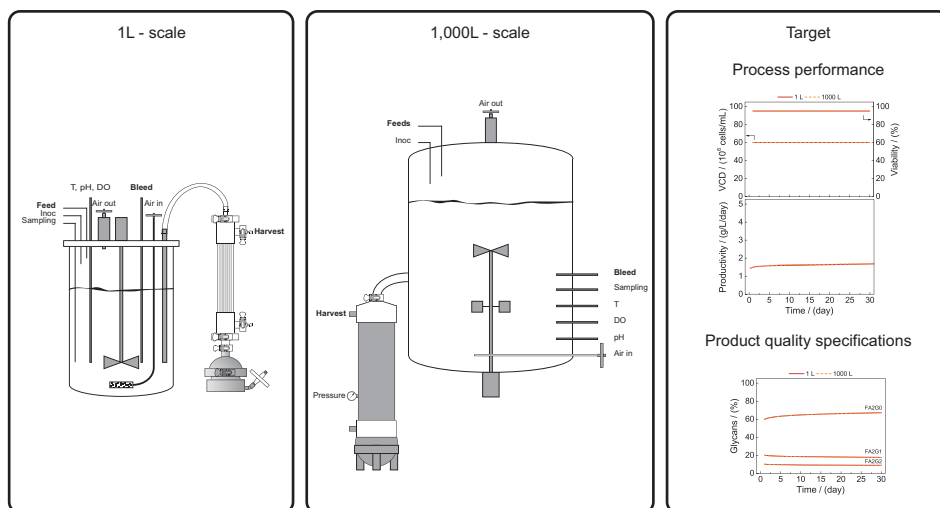


Figure 5.1 Illustration of the scale-up of a perfusion process: The process design and development end at the litre scale, and, when further increasing the scale, process conditions and equipment have to be scaled up to the commercial scale (1,000 L). Ideally, process performance (e.g., viable cell density, viability, productivity) and product quality attributes (e.g., N-linked glycosylation patterns) should remain unchanged.

different rates of oxygen uptake and carbon dioxide removal (Amanullah et al. 2001, Sieblist et al. 2011). From the point of view of a cell travelling in such a heterogeneous space, this translates into being exposed to continuously changing conditions in terms of pH, oxygen concentration and mechanical stress, which obviously affects its metabolism. In particular, in large-volume stirred vessels, the cells move in time from regions at very high shear – close to the impeller – to much calmer regions – for example, closer to the reactor ceiling – where the shear is lower. This leads to a kind of oscillatory environment, between high and low stress, which can significantly affect the cell behaviour (Neunstoecklin et al. 2015, Sieck et al. 2013).

Besides stirred vessels, which are quite common in chemical engineering operations, cell retention devices are unique and offer specific and relevant challenges. In particular, the scale-up issue results not only from the cell retention device itself but from the entire loop, where the device is inserted, which includes a pump, in the first place, but also various pipes and valves where the cells are exposed to a different and largely uncontrolled environment. In general, this may provide different types of additional stresses to the cell, related, for example, to poor temperature control, lack of oxygen or fluidynamic and mechanical conditions due to the loop geometry and the presence of an external pump. As a result, the entire design of the loop around the retention device, including the cell's residence time, becomes a critical component of the bioreactor scale-up (Voisard et al. 2003).

The issues just described become particularly challenging due to the very high cell densities in perfusion systems, which inevitably result in elevated overall production and consumption rates of metabolites and nutrients (Goudar et al. 2011). Nutrient supply and gas-liquid mass transfer may easily become limiting for the reactor operation (Mercille et al. 2000). In order to maintain the desired dissolved oxygen level in the bioreactor, the oxygen transfer rate (OTR) has to balance the cellular oxygen uptake rate (OUR). This requires a sufficiently high oxygen rate mass transfer

coefficient k_La – say above 5 h^{-1} as a rule of thumb – which strongly depends on sparger type and geometry, impeller configuration, volumetric gas flowrate and agitation rate (Godoy Silva et al. 2010, Ozturk 1996). Oxygen consumption is strictly connected to the production of the undesired natural byproduct, carbon dioxide (CO_2), which is a result of the cellular metabolism and typically produced in similar molar amounts by mammalian cells (Goudar et al. 2007, Gray et al. 1996). The produced carbon dioxide affects the reactor pH, which is usually controlled through a suitable buffer system and affected by the mixing conditions. This results in a complex environment with specific levels of pH, pCO_2 and osmolality that need to be kept under close control since they might negatively affect cellular growth and productivity and even impact product quality, particularly when the CO_2 level increases above about 100 mmHg (Brunner et al. 2018, Gray et al. 1996, Jiang et al. 2018, Zhu et al. 2005). Higher cell densities also imply more viscous suspension cultures, which necessitate higher agitation energies to remove heterogeneities in the metabolite and gaseous species composition within the bioreactor (Ozturk 1996).

The challenges described in the preceding paragraph indicate the need for good mixing in order to make the entire cell suspension sufficiently uniform, as well as to satisfactorily deliver nutrients and exchange gaseous species. This has to comply with practical limitations coming from the stress levels that harm cell viability and a reliable transfer of the process across scales. For this, it is necessary to develop proper scale-down procedures to identify maximum operating ranges for scale-independent parameters and to mimic large-scale conditions to evaluate possible adverse effects on the other (Neunstoecklin et al. 2015). Such parameters include energy dissipation rates, mixing times and gas–liquid mass-transfer rates, which are further discussed in Section 5.2.

Before entering the details of the effect of scale on the reactor behaviour, we have to at least briefly mention some peculiarities of today's biomanufacturing, which will greatly impact the structure and design of commercial production sites in the near future (Bielser et al. 2018, Croughan et al. 2015, Karst et al. 2018, Shukla & Gottschalk 2013, Thomas 2017, Walther et al. 2016). The way we produce therapeutic proteins is at a stage of rapid evolution driven by cost pressure, access to developing markets, growing competition and clinical pipelines with new and more complex molecular formats, as well as limited manufacturing capacity, increasing regulations concerning product quality and process knowledge (such as the quality by design initiative), and the need for a higher degree of flexibility in manufacturing (Klutz et al. 2016, Siganporia et al. 2014, Walther et al. 2015, Yu et al. 2015). Process intensification through continuous and integrated operation at various stages of the production process could be the answer to some of these challenges and concerns. Continuous operation, particularly at the fermentation level, has been so far limited to the production of fragile therapeutic proteins (such as Factor VIII), and has not yet been applied in more general terms (Bödeker et al. 2013, Langer 2011). However, the possibility to integrate the different unit operations within a single end-to-end continuous production process has now been demonstrated, and its use is recommended in a number of possible production scenarios. This represents an attractive solution to downsize manufacturing units and envision more flexible production platforms (Karst et al. 2018, Klutz et al. 2015, Pollock et al. 2017, Whitford 2014). With the reduction of the equipment size, single-use applications are becoming more and more of

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interest. Across the industry, experts envision the manufacturing plant of the future to be fully single-use, closed and continuously operated within a modular ballroom facility, without today's large hold-up tanks (Klutetz et al. 2015, Shukla & Gottschalk 2013, Whitford 2014). This obviously would not be the solution for all production situations, and many different variations will be worth considering, including hybrid technologies, with either a batch upstream process or a batch polishing unit, and only single units operated in continuous mode (Klutetz et al. 2016). We come back to these aspects towards the end of this chapter to complete the picture of current and future commercial-scale manufacturing in the biopharmaceutical industry.

5.2 SCALE-UP OF STIRRED BIOREACTORS

A continuous perfusion culture typically uses a stirred tank bioreactor coupled with an internal or external cell separation device. While discussing the scalability of the cell separation device in Section 5.6, this section focuses on the scale-up of the stirred vessel. Mammalian cell cultures require specific process design and equipment layout due to their unique characteristics, mainly related to a high sensitivity to shear, to (non-physiological) carbon dioxide concentrations and to variations in the environmental conditions (Chu & Robinson 2001, Nienow 2006). In the following, we discuss critical scale-up conditions – that is, appropriate mixing, sufficient oxygen transfer and adequate stripping of carbon dioxide at low shear rate (Chisti 2001, Godoy Silva et al. 2010, Nienow 2006, Sieblitz et al. 2011). We describe different approaches to combine the characterisation of the process equipment with the process knowledge in order to determine the optimal operating ranges for the perfusion culture.

Traditional scale-up strategies are rather based on empirical experience than on quantitative scientific considerations. Typically, scale-dependent parameters, such as stirring speed and gas flowrate, are adjusted at each scale during the scale-up process (Eon-Duval et al. 2012, Rouiller et al. 2012). This is a very labor-intensive approach since it requires proper ad hoc experimental activity at each scale.

On the other hand, scale-independent parameters can be defined within a reactor scale-down model and then implemented during scale up. This includes parameters such as maximum hydrodynamic stress, minimum oxygen supply and carbon dioxide removal rates, which can be used to define cell dependent ranges of operating conditions for each scale (Eon-Duval et al. 2013, Neunstoecklin et al. 2015, Sieck et al. 2014). Agitation speed and sparging intensity are valuable tools for correcting scale-up issues. However, it is necessary to address these issues systematically by performing small-scale studies that aim at simulating large-scale conditions for the investigation of possible adverse effects on cell behaviour. With respect to this, geometric similarities across different scales can provide applicable engineering rules for the determination of relevant parameters – such as the average energy dissipation rate, mixing time and gas mass-transfer rates. However, in practice, the bioreactors across the biopharmaceutical industry are often rather diverse, and this hampers the application of simple geometric scaling rules and requires either experimental measurements or computational estimation of the critical scale-up parameters (Eibl et al. 2010, Nienow et al. 2013, Villiger et al. 2018).

5.2.1 Aeration

Oxygen is a key substrate in mammalian cell cultures, particularly in the perfusion mode where the high cell densities need very large oxygen transfer rates. The low solubility of oxygen in water-based culture media makes a continuous oxygen supply necessary. The rate of oxygen supply has to balance the rate of cell oxygen consumption; otherwise, the dissolved oxygen (DO) concentration may fall below critical values – in general, in the order of 1 to 10 per cent of air saturation – resulting in changes in the cell's metabolism, which negatively impact product quality and quantity (Ozturk & Palsson 1990). However, elevated oxygen concentrations (e.g., in the case of pure oxygen sparging) may stimulate the generation of reactive oxygen species, which can alter cellular metabolism or cause cell death, as discussed in Chapters 2 and 3 (Hsieh et al. 1986). For this, it is necessary to maintain cell cultures within an optimal range of DO that typically goes from 40 to 60 per cent of air saturation (Miller et al. 1987, Ozturk & Palsson 1990).

In general, animal cell lines under favorable oxygen supply conditions exhibit cell-specific oxygen consumption rates in the range of 0.05 to 0.5×10^{-12} mol/cell/h (Fleischaker & Sinskey 1981). Considering a targeted viable cell density of $1.0 - 2.0 \times 10^8$ cells/mL, the total oxygen uptake rate in a perfusion culture ranges between 5 and 100 mol/m³/h, which is not too different from yeast or bacteria cultures. This means that mammalian cell perfusion bioreactors require high values of the oxygen transfer rate (OTR) given by the product of the volumetric mass transfer coefficient, $k_L a$ [h⁻¹], and the driving force, ΔC_{O_2} .

$$\text{OTR} = k_L a \Delta C_{O_2} = k_L a (C_{O_2}^* - C_{O_2}), \quad (5.1)$$

with $C_{O_2}^*$ representing the saturated DO concentration in the cell culture – that is, the value in equilibrium with the oxygen concentration in the gas phase – and C_{O_2} representing the actual DO concentration in the liquid phase. The maximum value of $C_{O_2}^*$ is 1 mM, corresponding to the case when pure oxygen at 1 atm is sparged into the bioreactor. The typical DO concentration in a bioreactor operation is about 50 per cent of air saturation, which corresponds to a concentration of 0.1 mM. Considering the OTR values needed in mammalian perfusion bioreactors mentioned earlier, it is seen that the mass transfer coefficient values have to be in the range of 5 up to 100 h⁻¹.

The value of $k_L a$ depends on several different factors, but mostly on the superficial velocity of the gas, V_S , and the average total energy dissipation rate ($\bar{\epsilon}_T$)_g, transferred to the liquid both through the stirrer, ($\bar{\epsilon}_T$)_{lg}, and the gas sparger ($\bar{\epsilon}_T$)_s. While the energy dissipated by the first one is defined by the mixer rotation speed and geometry, the energy delivered by the second is mainly affected by the liquid density and the superficial gas velocity, which is defined as the ratio between the volumetric gas flowrate, Q_g , and the bioreactor cross section, A :

$$V_S = \frac{Q_g}{A}. \quad (5.2)$$

In general, with respect to the energy dissipation rate, the contribution of gas sparging is minor since low air flowrates are applied to minimise cell damage by bubble bursting. The energy input is, in fact, dominated by the stirrer under gassed conditions and is typically a function of the stirrer rotation speed and diameter, the culture density and the liquid volume inside the reactor, as discussed in Section 5.2.3.

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Despite other aeration techniques – such as membrane aeration, surface aeration or the use of oxygen carriers or perfluorocarbons – bubble aeration, also referred to as sparging, is not only the most common way to introduce and remove gaseous nutrients and volatile by-products from the cell culture, but it is also the one that enables the highest transfer rates (Godoy Silva et al. 2010, Jesus & Wurm 2011, Ozturk 1996). The selection of a proper sparger device represents a compromise between maximising gas transfer rates (oxygen supply and carbon dioxide removal) while minimising cell damage and foam formation. Open pipe spargers, ring sparger and micro or frit spargers are the most commonly used aeration devices. The first one is simply a pipe ending in a nozzle through which air is introduced into the vessel, resulting in a moderate control of the bubble size, through the nozzle, diameter and position, and the gas flowrate. However, this device typically produces large bubbles, and it is therefore referred to as a macrosparger. The advantages of such devices are minimised foaming and cell death, but large bubbles have a low surface-per-unit-volume ratio and, therefore, do not support high mass-transfer rates, thus requiring high gas volumetric flowrates in order to guarantee sufficient oxygen supply. A more advanced configuration is the ring sparger, constituted by a tube with a ring shape and a number of holes of defined diameter, typically around 1 mm, allowing the simultaneous generation of multiple bubbles. On the other hand, in a microsparger, porous materials are used to create a large amount of very small bubbles in the micrometer range. Small bubbles are ideal for mass transfer because of the larger specific gas–liquid interphase surface area. However, the use of frit spargers results in reduced carbon dioxide stripping and more foam formation. Very often, a reasonable solution is to combine a frit and a large bubble sparger and use antifoaming chemicals to prevent the negative impact of foaming. In addition, shear protectants, such as Pluronic F68, can prevent cell damage caused by bubble bursting at the liquid surface (Sieblist et al. 2013). The prevention of foam formation, by antifoam addition, on the other hand, may lead to a decrease in the mass-transfer rates (Lavery & Nienow 1987).

5.2.2 Mixing

Mixing in bioreactors is needed to uniformly distribute cells, nutrients and gas bubbles – as well as the added liquids, such as base or nutritional feeds – in order to avoid spatial variations or gradients within the bioreactor, which may harm the cells. Typically, this is accomplished through the addition of energy to the cell culture broth through both mechanical agitation and gas sparging. In the ideal case, such energy should be delivered to the culture uniformly so that

1. Every single cell is exposed to a similar mixing intensity although changing in time, as seen earlier.
2. Nutrients, metabolites and by-products are uniformly distributed throughout the reactor so that the cells do not experience any variation and keep their metabolism unchanged (Godoy Silva et al. 2010, Sieblist et al. 2011).

From these considerations, it is seen that a proper design of the mixing system is needed at all scales in order to provide proper cell culture conditions while maintaining cell viability. An inadequate mixing design can, for example, allow cells to settle at the bottom of the reactor, thus creating zones with limited mass and energy transfer and a quite different environment. On the other hand, aggressive mixing is also

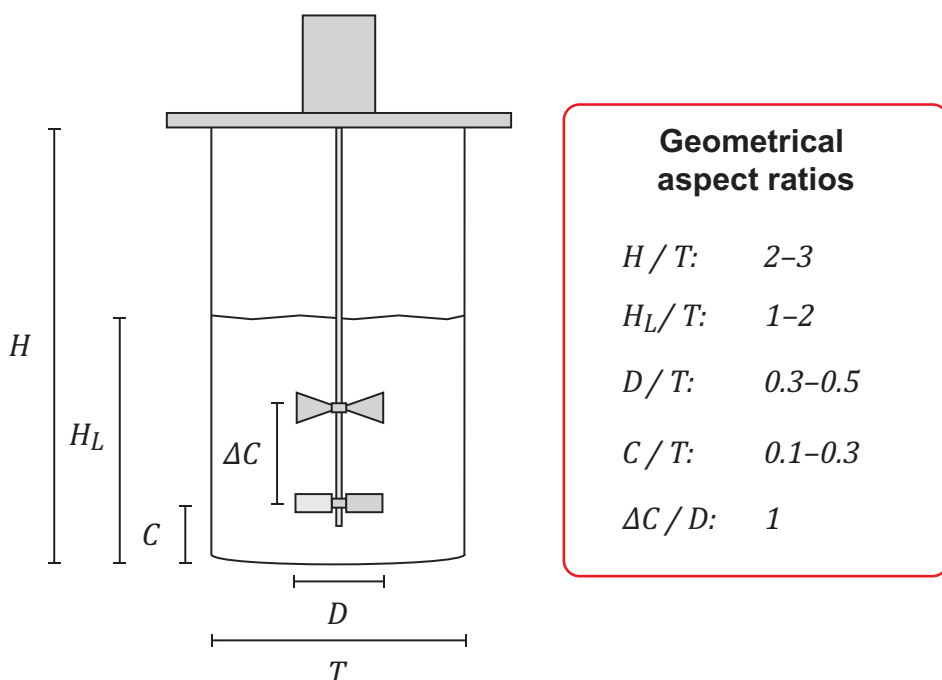


Figure 5.2 Standard geometrical representation of a bioreactor set-up with the most important dimensions and aspect ratios, which have to be considered during the vessel scale-up.

negatively impacting the cells, as the resulting shear stress experienced by the cells can cause cell-growth inhibition, decrease the specific productivity or induce cell death (Marks 2003). Process and chemical engineering understanding of reactor systems with regard to mixing and power input – coupled, if needed, with the application of computational tools – are requested for smooth technology transfer across scales.

In biotech and biopharma applications, the reactor system of choice is the classical stirred vessel at all scales: from the laboratory up to the large commercial scale. Mixing within such systems is strongly affected by the vessel geometry, the presence of baffles of various shapes, and the geometry and rotation speed of the stirrer. All of this leads to different energy-dissipation rates and mixing characteristic times.

In terms of geometry (See Figure 5.2), a bioreactor can be seen as a common stirred vessel, characterised by proper dimensionless aspect ratios that have to be kept constant when aiming at geometric similarity in the reactor scale-up (Godoy Silva et al. 2010). Typical geometries include the aspect ratio of tank height (H) to tank diameter (T), equal to approximately $\frac{H}{T} = 2 - 3$. In the centre, a shaft penetrates the tank, from either the top or the bottom, connected to a motor at the outside end, while in the inside, one or more stirrers are attached to the same shaft. In the most common configurations, the aspect ratio of the impeller diameter (D) to the tank diameter is between $\frac{D}{T} = 0.3 - 0.5$. The filling height of the cell culture broth (H_L) should be around $0.7H$, resulting in an aspect ratio of filling height and tank diameter between $H_L/T = 1.0 - 2.0$. To determine the location of the stirrer, a distance from the bottom of the tank equal to $C = 0.1 - 0.3T$ appears to be a reasonable choice. In

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the case that more than one impeller is used in the vessel, the distance between them (ΔC) can be highly variable but is typically around one stirrer diameter.

Besides the reactor geometry, the choice of the stirrer(s) represents an important step in the reactor design, since various shapes are available leading to quite different mixing characteristics (Nienow 2010). Common designs include radial-flow stirrers such as the Rushton turbine (RT), providing a high level of intense mixing with high local shear rates, and a prevailing radial flow pattern, as well as down-pumping axial-flow stirrers such as the marine propeller, which result in an improved axial bulk movement with respect to the RT. Axial flow or pitched-blade turbines produce a flow pattern somewhere in between the others. The stirrer selection depends on several considerations, but, in particular, its specific power number (P_0), which is a function of the stirrer type itself as well as its relative size (D/T) and position in the vessel (C/T). In the case of an impeller, the power number can be used to calculate the energy dissipated into the system, P , which upon normalisation with respect to the reactor volume leads to the specific energy dissipation rate, $\bar{\epsilon}_T$.

Comparing radial-flow and axial-flow impellers, it is seen that the first ones are typically characterised by a high specific power number and consequently referred to as *high shear impellers*, while the second ones have comparably lower specific power numbers and are consequently referred to as *low/intermediate shear impellers*. Due to the shear sensitivity of animal cells, the *low/intermediate shear impellers* are often preferred for these cultures. Nevertheless, *high shear impellers*, such as the Rushton turbine, allow the breakage and complete dispersion of gas bubbles at reasonably low impeller speeds and are therefore often used at bench-scale while, e.g., (*low/intermediate shear*) marine propellers are favoured at the manufacturing scale (Nienow 2010). For all impeller types, the maximum local specific energy dissipation rate measured close to the impeller is relatively high compared to the average energy dissipation rate in the bulk, which is what we refer to as mixing heterogeneity and constitutes a crucial aspect in reactor scale-up.

In large-scale units, multiple impellers are often attached to a single-stirrer shaft so as to optimise the overall mixing performance. A typical recommended distance between two of such impellers is between one to two times the impeller diameter so that the flow patterns of the two impellers do not interfere with each other (Godoy Silva et al. 2010). Mixing effectiveness can be further improved by the implementation of baffles, which are placed perpendicular to the tank wall at 90° increments around the vessel. In the presence of baffles, large swirls can be broken, thus achieving better mixing conditions.

5.2.3 Estimation of Mixing and Mass Transfer Characteristic Times

In order to evaluate the effectiveness or characteristic time of mixing and gas–liquid mass transfer in a bioreactor, an a priori physical characterisation is necessary, which is independent of the scale and the operating mode, batch or continuous, of the bioreactor. First, we focus on the mixing effectiveness of a given set-up with a given impeller and sparging system (Karst et al. 2016, Nienow 1997). The power input (P) can be estimated experimentally as the product between the measured torque on the impeller shaft (M) and the applied agitation speed (N):

$$P = 2\pi MN, \quad (5.3)$$

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from which the average energy dissipation rate is readily obtained as follows:

$$\bar{\epsilon}_T = \frac{P}{V_R}. \quad (5.4)$$

Alternatively, one can estimate the power input from the definition of the dimensionless power number P_0 under unaerated conditions, defined as

$$P_0 = \frac{P}{\rho N^3 D^5} \quad (5.5)$$

as follows:

$$P = P_0 \rho N^3 D^5, \quad (5.6)$$

where P_0 can be computed from semi-empirical correlations (Bates et al. 1966) and provides the power number for specific stirred vessel configurations as a function of the Reynolds impeller number, Re_{imp} . For example, under turbulent conditions ($Re > 2 \times 10^4$, where $Re = \frac{\rho_L u L}{\mu_L}$ with u the fluid velocity, and L the characteristic length), the power number is independent of the Reynolds number in the case of a Rushton impeller with a power number value of about 5.0 (*high shear*), while for a classical six-blade, 45° pitch turbine a value of about 1.7 is reported (*intermediate shear*) (Bujalski et al. 1987, Nienow 2010, Rutherford et al. 1996). Therefore, we can estimate the energy dissipation rate for a given impeller configuration and its specific power number as follows:

$$\bar{\epsilon}_T = \frac{P_0 \rho N^3 D^5}{V_R} \quad (5.7)$$

In order to evaluate the effect of sparging on mixing, this analysis has to be repeated in aerated conditions in order to determine the gaseous power input, P_g , which often results in a decrease of the amount of power the impeller is able to dissipate into the liquid – in particular, for high-shear conditions. This is due to a process referred to as *flooding*, consisting in the generation of air cavities around the impeller, which are stabilised by the lower pressure values prevailing in this region. Semi-empirical correlations for the estimation of the power input under aerated conditions are reported, in the case of flat-blade turbines, by Hughmark (1980).

$$\frac{P_g}{P} = 0.10 \left(\frac{Q_g}{NV_R} \right)^{-\frac{1}{4}} \left(\frac{N^2 D^4}{g D_i V_R^{\frac{2}{3}}} \right)^{-\frac{1}{5}}, \quad (5.8)$$

which has been derived for the following ranges of operating conditions:

$$\begin{aligned} 0.87 < \rho_L < 1.6 & \quad \left[\frac{\text{kg}}{\text{m}^3} \right] \\ 0.0008 < \mu_L < 0.028 & \quad [\text{Pa} \times \text{s}] \\ 0.025 < \sigma_L < 0.072 & \quad \left[\frac{\text{N}}{\text{m}} \right] \\ 0.1 < T < 1 & \quad [\text{m}] \\ 0.25 < \frac{D}{T} < 0.46 & \quad [-] \end{aligned}$$

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$$0.31 < \frac{P_g}{P} < 0.8 \quad [-],$$

with the volumetric gas flowrate, Q_g , the agitation speed N , the bioreactor volume V_R , the impeller diameter D , the impeller blade width D_i and the acceleration of gravity g .

The time scale of mixing is inversely proportional to the turbulent intensity. In particular, the mixing characteristic time, θ_m , can be estimated from the two following expressions (Ma et al. 2003, Nienow 1997): the first one applies when the aspect ratio of the liquid tank height to the tank diameter ($\frac{H_L}{T}$) is equal to 1 and the second when the liquid tank height is larger than the tank diameter, so that $H_L > T$:

$$\theta_m = 5.3 \left(\frac{1}{N} \right) \left(\frac{1}{P_0^{\frac{1}{3}}} \right) \left(\frac{T}{D} \right)^2 \quad (5.9)$$

$$\theta_m = 3.3 \left(\frac{1}{N} \right) \left(\frac{1}{P_0^{\frac{1}{3}}} \right) \left(\frac{T}{D} \right)^{2.43}. \quad (5.10)$$

In order to keep constant the mixing times during scale-up, the product of $\left(\frac{1}{N} \right) \left(\frac{1}{P_0^{\frac{1}{3}}} \right)$ should be kept constant as $\frac{T}{D}$ is kept constant by definition:

$$\left(\frac{1}{N} \right) \left(\frac{1}{P_0^{\frac{1}{3}}} \right) = \text{constant}, \quad (5.11)$$

which, using Equation (5.6), leads to

$$\left(\frac{1}{N} \right) \left(\frac{\rho N D^{\frac{5}{3}}}{P^{\frac{1}{3}}} \right) = \left(\frac{\rho D^{\frac{5}{3}}}{P^{\frac{1}{3}}} \right) = \text{constant}. \quad (5.12)$$

This equation indicates that during scale-up of geometrically similar tanks (i.e., constant ratio $\frac{T}{D}$), the mixing time can be maintained constant only if the power input per unit volume is increased proportionally to the square of the impeller diameter D . However, due to hydrodynamic concerns at the large scale (as we will discuss later), these reactors tend to be designed so that the power input per unit volume is constant or decreases with the scale.

The characteristic mixing time can be measured experimentally through the dye decolorisation method, which can be applied in transparent vessels (Sieblist et al. 2016). Initially, an amount equal to 0.2 per cent of the reactor volume of a 1 weight per cent starch solution is added to the reactor, and then a 1 M iodine–potassium iodide solution is added in order to color the liquid inside the reactor. After this, a given amount of 1 M sodium thiosulphate solution is added, which reduces the iodine leading to the decolorisation of the liquid. The time between the addition of the sodium thiosulphate solution and the complete decolorisation provides an approximate estimate of the characteristic mixing time. Due to GMP regulations, this method is not applicable in production vessels.

Villiger et al. (2018) used pH-shift experiments in order to experimentally determine the characteristic mixing time. For this, they carried out pulse additions of concentrated NaOH solutions (32 per cent, 10.8 mol/L) with a volume of 0.1 per cent of the bioreactor volume. The pH was monitored every 5 seconds with two pH probes – one located at the bottom and the other one at the top of the reactor.

The starting point of the measurement was aligned with the addition of base, and the mixing characteristic time was taken as the point when both probes reached their new and equal pH value. This experiment can be repeated several times to obtain a reliable estimate of the mixing time, and, in addition, by varying agitation and aeration conditions, their effect on the mixing characteristic time can be evaluated.

Alternatively, computational fluid dynamic (CFD) approaches can be used in order to mimic *in silico* the same experiments described and compute the relative characteristic mixing times based on simulations, as discussed later in Section 5.4.

In addition to mixing effectiveness, the evaluation of gas–liquid mass transfer is of substantial importance not only to secure sufficient oxygen supply but also to remove carbon dioxide, at the different reactor scales. This is measured by the volumetric gas–liquid mass transfer coefficient k_La defined by Equation (5.1), which depends on many of the stirred vessel elements: stirrer, sparger, baffles, liquid volume and the relative operational settings such as, for example, stirring speed and gas flowrate. We have seen in the previous section that in order to support perfusion cultures with cell densities above 10^8 cells/mL, a minimum of k_La of 5 h^{-1} is necessary, but higher values are definitely recommended (Ozturk 1996). For a given set-up, the corresponding k_La can be measured using the dynamic gassing method (Van't Riet 1979). First, the filled bioreactor is stripped out of all oxygen and carbon dioxide through nitrogen aeration. Next, the desired operating conditions are applied in terms of gas flowrate, agitation speed and oxygen concentration in the gas stream. By changing such conditions – for example, through a factorial design scheme – the k_La values can be calculated by fitting the measured oxygen concentration values as a function of time, as described by Bandyopadhyay et al. (1967). The obtained values are then fitted as a function of the prevailing operating conditions through the classical Van't Riet equation (1979):

$$k_La = K \left(\frac{P_g}{V_R} \right)^\alpha V_S^\beta, \quad (5.13)$$

where, for pure water,

$$K = 0.026 \quad \alpha = 0.4 \quad \beta = 0.5$$

and, for ionic solutions,

$$K = 0.002 \quad \alpha = 0.7 \quad \beta = 0.2.$$

This equation has been derived for the following operating conditions:

$$\begin{aligned} 2 \times 10^3 < V_R < 4.4 & \quad [\text{m}^3] \\ 500 < \frac{P}{V_R} < 10,000 & \quad \left[\frac{\text{W}}{\text{m}^3} \right], \end{aligned}$$

with P_g denoting the gaseous power dissipated by the stirrer, V_R the volume of the bioreactor and V_S the superficial gas velocity defined by Equation (5.2). Alternatively, the expression reported by Chandrasekharan & Calderbank (1981) can be used:

$$k_La = \frac{E}{T^4} \left(\frac{P_g}{V_R} \right)^A Q_q^{\frac{A}{\sqrt{T}}}, \quad (5.14)$$

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with

$$E = 0.0248 \quad A = 0.551.$$

This equation has been derived for the following operating conditions:

$$\begin{aligned} 90 < \frac{P}{V_R} < 1700 & \quad \left[\frac{\text{W}}{\text{m}^3} \right] \\ 2.6 \times 10^{-3} < V_S < 1.8 \times 10^{-2} & \quad \left[\frac{\text{m}}{\text{s}} \right] \\ 0.3 < T < 1.2 & \quad [\text{m}], \end{aligned}$$

with the tank diameter T , the reactor volume V_R , the gaseous power input P_g , the volumetric gas flowrate Q_g and the fitting parameters A and E .

As for the mixing effectiveness, computational fluid dynamics can also be applied to determine the local k_{La} values based on stirred vessel simulations, as discussed in the next section.

In general, when considering both mixing and aeration, it is important to find a maximum scale-independent variable, such as the specific energy dissipation rate, which provides sufficiently high mixing effectiveness and gas–liquid mass-transfer rates. However, such conditions should also not be harmful to the cells with respect to their shear sensitivity. Therefore, it is necessary to further analyse and evaluate the reactor fluid dynamic conditions (Nienow 1998) with respect to the shear rates to which cells are exposed at different scales.

5.3 HYDRODYNAMIC STRESS

As previously discussed, energy needs to be dissipated into a bioreactor to achieve satisfactory conditions in terms of mixing and mass transfer. However, this inevitably creates shear stresses, and we need to make sure that these are not harmful to the cells. In the frame of process scale-up, it is convenient to consider these aspects in terms of scale-independent variables, such as mass-transfer rate and overall energy dissipation rate, rather than scale-dependent variables, such as stirring speed and sparging rate (Eon-Duval et al. 2013, Neunstoecklin et al. 2015, Sieck et al. 2014). In fact, we expect to change the second ones during scale-up in such a way to keep the first ones unchanged.

In this section, we discuss the impact of shear stresses on cell viability. In general, mammalian cells show proper resistance against hydrodynamic stress, and the maximum tolerable stress threshold is highly cell line dependent (Chisti 2001, Neunstoecklin et al. 2015, Tanzeglock et al. 2009). For this, it is convenient to have a general procedure that can be used to determine such maximum tolerable threshold value for any given cell line and for any given experimental set-up. From a regulatory perspective, this is even more important and is aligned with the Quality-by-Design approach. It is fundamental to have similar hydrodynamics in bench- and large-scale manufacturing equipment to achieve similar process performance.

It is indeed difficult to determine the maximum hydrodynamic stress in multi-phase systems – in particular, in bioreactor systems. Computational fluid dynamics represent an option, but in the presence of multiple phases and complex geometries, its application is not straightforward. Villiger et al. (2015) developed a simple

experimental approach that can be applied to any stirred vessel at any scale and, actually, also to units of any other type, like pumps, filters or others. It is based on the use of a suspension of poly(methyl methacrylate) (PMMA) nanoparticles synthesised by monomer-starved semi-batch emulsion polymerisation (Sajjadi & Yianneskis 2003). The initial suspension is aggregated in a vessel by stirring and adding a well-defined amount of sodium chloride solution, above the critical coagulation concentration (CCC), to destabilise the suspension and promote aggregation of the nanoparticles and formation of aggregates (Moussa et al. 2007). Such aggregates of well-defined size are then diluted with deionised water to prevent any further aggregation and to obtain a stable aggregate suspension. Now, the idea is that these aggregates are fragile and, if exposed to a certain shear stress, can break, and actually, the higher the stress they are exposed to, the smaller the size they reach. It follows that such suspensions can be introduced in any reactor or device of other sort and circulated for a sufficiently long time so that breakage can occur until steady-state dimensions are reached. Their final size can then be measured, for example, by small-angle light scattering (SALS), and this provides a measurement of the maximum shear rate that the aggregates have experienced inside the bioreactor. Clearly, this does not tell us anything about where in the unit such maximum shear occurs, but only what is the maximum shear experienced by the aggregates (or, equivalently, by the cells) somewhere in the device. To obtain a quantitative estimate of this, we obviously need a calibration curve to go from the measured aggregate size to the value of the maximum shear experienced, which Moussa et al. (2007) provided using a dedicated device. In particular, they used the axisymmetric extensional flow generated at the entrance of contracting nozzles of various diameters and flowrates (Saha et al. 2014). The corresponding shear rate was computed by computational fluid dynamics (Soos et al. 2010). Thus, by correlating the computed shear values with the aggregate sizes measured in the same device when circulating the PMMA aggregate suspension, the desired calibration curve could be constructed. This technique was used in a series of experiments performed in classical stirred tank reactors with given tank diameter, Rushton turbine impeller and a four-element baffle configuration, for various impeller rotation speeds and air flowrates. The obtained values of the maximum effective hydrodynamic stress, τ_{max} , for a given stirred tank reactor at given operating conditions were in good agreement with literature data (Neunstoecklin et al. 2015, Villiger et al. 2015).

Neunstoecklin et al. (2016) used this experimental approach to determine the relative importance of stirring and sparging to the maximum hydrodynamic stress in a 300 L bioreactor. For this, τ_{max} was measured for various combinations of stirring and sparging systems. In particular, two types of ring spargers, located below a Rushton impeller, were considered: one with eight sintered spargers (average pore diameter equal to 50 μm) and the other with eight nozzles with a diameter of 1 mm. The obtained results are shown in Figure 5.3 for different sparging flowrates as a function of the stirring speed, represented in terms of the Reynolds number, defined as $Re_{imp} = \frac{\rho_L N D^2}{\mu_L}$. It is seen that the maximum hydrodynamic stress for non-aerated conditions increases linearly with increasing Re_{imp} , in a log–log scale. When considering low gas flowrates (i.e., 0.03 vvm) the measured values of τ_{max} when using the sintered sparger are comparable to those measured for single-phase conditions, indicating the negligible contribution of small bubbles. In contrast,

5.3 Hydrodynamic Stress

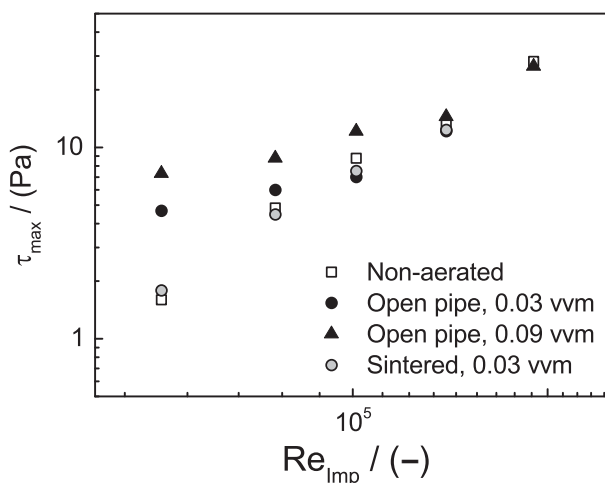


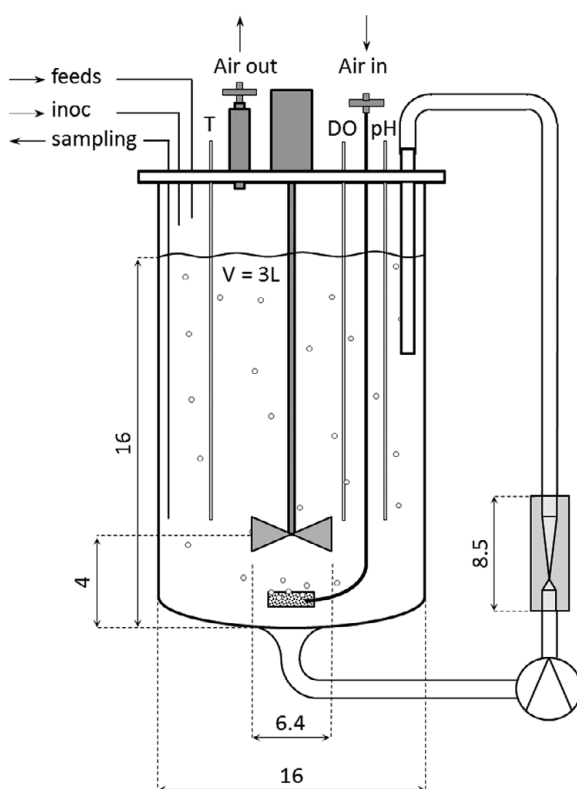
Figure 5.3 Maximum hydrodynamic stress in a 300 L bioreactor measured by the shear-sensitive PMMA system developed by Villiger et al. (2015). Two types of spargers, one with 1 mm open nozzles and the other with a 50- μ m sinter sparger, were investigated at various agitation speeds and gas flowrates. Open squares correspond to the non-aerated single-phase condition. The applied gas flowrates for the nozzle sparger were equal to 0.03 vvm (filled circle) and 0.09 vvm (filled triangle), while for the 50- μ m sparger, the gas flowrate was equal to 0.03 vvm (grey circle). © Reprinted by permission from Springer Nature, *Applied Microbiology and Biotechnology*, ‘Pilot-scale verification of maximum tolerable hydrodynamic stress for mammalian cell culture’, Benjamin Neunstoecklin, Thomas K. Villiger, Eric Lucas, Matthieu Stettler, Hervé Broly, Massimo Morbidelli and Miroslav Soos, 2016

when using the 1 mm open pipe system, which generates larger bubbles, the value of τ_{max} increases at low Re_{imp} to about 5 Pa with gas flowrates of 0.03 vvm and even more, at about 8 Pa, for larger gas flowrates of 0.09 vvm. By increasing the impeller speed, the value of τ_{max} increases in all cases, but the effect of sparging decreases and becomes negligible at sufficiently large values of Re_{imp} where the maximum hydrodynamic stress is dominated by the turbulence generated by the impeller (Neunstoecklin et al. 2016).

Neunstoecklin et al. (2015) used this technique to determine the maximum operating hydrodynamic stress for mammalian cell cultures for different cell lines (CHO and Sp2/0). The authors realised a scale-down model to determine the threshold values of hydrodynamic stress, which was evaluated based on different cell characteristics – that is, cell viability, growth, morphology, metabolism and productivity. In particular, a modified bioreactor set-up was used in order to expose the cells to different and well-defined hydrodynamic stresses, as illustrated in Figure 5.4. In addition to the typical stirred vessel, this set-up is equipped with an external loop, driven by a contact-free centrifugal pump, containing a nozzle. Pumping the cell culture liquid through the external loop with the nozzle, an oscillating hydrodynamic stress is imposed to the cells, similar to what they experience in a large scale bioreactor, with the high stresses in the nozzle representing the regions in the vicinity of the stirrer and the lower stresses in the stirred vessel representing the bulk zone away from the stirred region (Soos et al. 2013). The exposure time in the loop was set to 90 seconds, which was correlated to the cells’ exposure time in zones with large hydrodynamic stresses

Figure 5.4 Set-up and geometry of the 3 L bioreactor system used by Neunstoecklin et al. (2015) to generate an oscillating hydrodynamic stress on the cells. Nozzles of various diameters were used for the generation of different shear rate maxima. All given values are in cm.

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in an in-house 5,000 L bioreactor. By varying the nozzle diameter, hydrodynamic stresses of different magnitudes can be imposed. Thus, by circulating the PMMA aggregate suspension inside the unit and measuring the corresponding equilibrium size, using the calibration curve, as mentioned before, the maximum shear experienced by the aggregates (and also by the cells, since they exhibit very similar size) inside the unit could be estimated.

The decrease in cell viability measured after 24 h of culture in bioreactors sparged with nozzles of different sizes are shown as a function of the maximum measured shear stress in Figure 5.5 for the two considered cell lines. It is seen that cell viability is not affected by shear stress values below a critical value, above which cell viability decreases.

This behaviour is common for both cell lines, with the maximum tolerable threshold value at τ_{max} of 32 ± 4 Pa for the CHO cell line and at 25 ± 2 Pa for the Sp2/0 cell line. Neunstoecklin et al. (2016) performed validation experiments for the given Sp2/0 cell line in a much larger 300 L pilot-scale bioreactor operated at different agitation and sparging rates to mimic hydrodynamic stress conditions in the range from 7 to 28 Pa and confirmed that, independent of the reactor scale, growth behaviour, metabolite concentrations, productivity and product quality showed a dependency on the different environmental stress conditions. Although this study was performed in fed-batch cultivations, similar concepts are also expected to apply in perfusion conditions. The maximum tolerable stress, τ_{max} , is, in fact, a cell-specific

5.4 Computational Fluid Dynamics

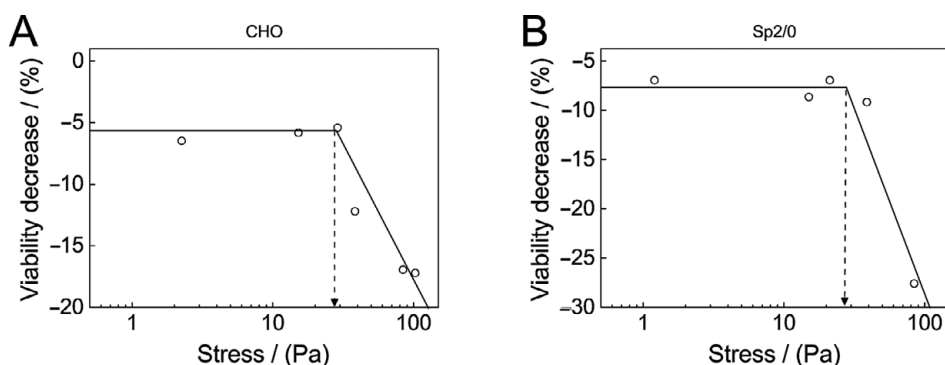


Figure 5.5 Example of the critical stress determination (dashed arrows) based on the viability drop during the first 24 h of the culture for a CHO (A) and a Sp2/0 cell line (B). © Reprinted from *Journal of Biotechnology*, 194, Benjamin Neunstoecklin, Matthieu Stettler, Thomas Solacroup, Hervé Broly, Massimo Morbidelli and Miroslav Soos, 'Determination of the maximum operating range of hydrodynamic stress in mammalian cell culture', 100–109, 2015, with permission from Elsevier

quantity and not a process-specific one, and therefore, it is expected to hold true for both batch and continuous operations and across different scales.

5.4 COMPUTATIONAL FLUID DYNAMICS

We have seen the importance of characterising bioreactors at different scales in terms of mixing, aeration and shear stress and discussed several experimental techniques to measure the corresponding relevant quantities. In this section, we briefly discuss the possibility of determining the same quantities through numerical simulations – that is, using computational fluid dynamics (CFD). The main advantage of CFD is to provide estimates of all local quantities in the entire reactor volume, thus providing a full description of the reactor heterogeneity, in time and space, which is particularly relevant for commercial scale bioreactors (Villiger et al. 2018). Computational fluid dynamics provides a valuable tool to evaluate this heterogeneity for various reactor systems of various geometries and sizes (Ranganathan & Sivaraman 2011) and at different operating conditions.

This methodology applies numerical techniques to solve the equations describing the motion of fluids within the bioreactor. The Navier–Stokes equations build the basis of CFD containing expressions for the conservation of mass, momentum and energy. The use of discretisation techniques enables the solving of the different equations for a finite set of volumes, so-called cells, and results in discrete flow fields. Assumptions, such as steady-state or inviscid flow, simplify the solutions and accelerate the convergence of the solutions (Sharma et al. 2011).

A typical CFD analysis includes three stages.

1. Building the model using appropriate meshing and boundary conditions, as well as information about the fluid properties.
2. Solving the various momentum, mass and energy balance equations by discretisation and producing results in some convenient form to be analysed.
3. Interpreting the data in the form of vectors and contour plots to extract the desired information (Versteeg & Malalasekera 1995).

In particular, the comparison of calculated CFD and experimentally determined values can be used for the validation of the obtained results. There is various commercial CFD software available for computational fluid dynamic simulations, which greatly simplify the last two steps (Sharma et al. 2011). This is relatively advanced software that allows one to select proper models for the bioreactor, simulate various geometries and the rotation of different impellers, and model the laminar or turbulent conditions inside the reactor. The software calculates the resulting flow field, with a bioreactor-size-dependent mesh size. Furthermore, it is possible to model pressure, momentum, turbulent kinetic energy and dissipation rates, thus computing the time and space distribution of hydrodynamic stresses. CFD can also be used to simulate the trajectory of a massless tracer, thus, for example, reproducing some of the mixing experiment described earlier and then comparing computed and experimental characteristic mixing times. Among the most common applications of CFD, we can list the determination of $k_L a$ values within different type of bioreactors, hydrodynamic forces within different bioreactor regimes, local energy dissipation rates and optimum impeller locations by estimating the different hydrodynamic properties and different flow regimes (Ahmed et al. 2010, Devi & Kumar 2017, Kelly 2008, Ranganathan & Sivaraman 2011, Soos et al. 2013). Of course, depending on the level of detail of the desired simulation and the complexity of the system in terms of number of phases, steady-state or transient behaviour, and overall size of the object to be simulated, the required computation time may grow significantly and become unfeasible. For this, various techniques to reduce the size of the process can be introduced – for example, using pseudo-phases or partitioning the entire reactor volume in separate zones. In principle, CFD can be successfully applied not only to gain fundamental understanding of different processes taking place in a given bioreactor configuration but also to optimise the operating conditions and bioreactor design during scale-up.

5.5 CHARACTERISATION OF THE BIOREACTOR: CASE STUDIES

In this section, we illustrate some case studies where the concepts discussed earlier have been applied to perform successful process scale-up from the bench to the manufacturing scale. In the first part, we discuss a comparative characterisation study of different reactor scales from the mL to the commercial scale in terms of mixing, aeration and hydrodynamic stress using both experimental and computational tools (Villiger et al. 2018). Next, using the results of an omics study by Bertrand et al. (2018), we compare cultures at different scales not only in terms of productivity and product quality but also at the level of intracellular processes. Finally, we describe the full characterisation of a perfusion set-up before proceeding with the discussion about the scale-up of cell retention devices in the next section.

Villiger et al. (2018) performed an experimental and computational comparative study of different animal cell culture bioreactors ranging from the micro- to the production scale. The study focused on the comparison of the fermenters and did not consider any auxiliary equipment, such as the cell retention device. The considered reactors are schematically illustrated in Figure 5.6 and include the disposable 15 mL ambr micro-bioreactor, two different 3 L scale glass vessel reactors (with different impellers) and the larger stainless-steel bioreactors with volume of 270 L, 5,000 L and 15,000 L. The comparison included the experimental determination of the

5.5 Characterisation of the Bioreactor: Case Studies

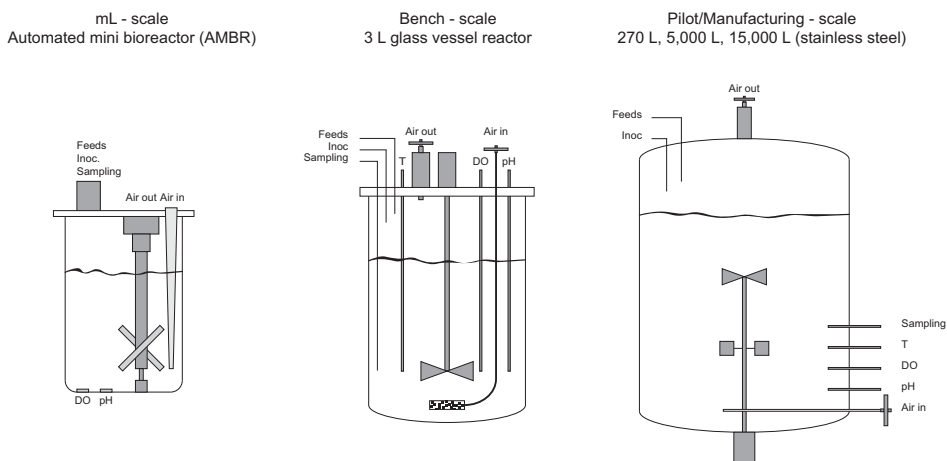


Figure 5.6 Schematic representation of the different reactor set-ups ranging from the mL-scale (ambr[®] 15 system) with nominal working volume of 10 mL, to the bench-scale with a nominal working volume of 3 L to the manufacturing scale with nominal working volumes of 270, 5,000, and 15,000 L. © Reprinted from *Biochemical Engineering Journal*, 131, Thomas K. Villiger, Benjamin Neunstoecklin, Daniel J. Karst, Eric Lucas, Matthieu Stettler, Hervé Broly, Massimo Morbidelli and Miroslav Soos, 'Experimental and CFD physical characterisation of animal cell bioreactors: From micro- to production scale', 2018, with permission from Elsevier

maximum hydrodynamic stresses, the mixing time, the volumetric gas–liquid mass transfer coefficient and the dissolved oxygen concentration at different locations in the reactor volume. The same quantities were also computed through computational fluid dynamic simulations in order to evaluate their heterogeneity in the reactor. Both experimental and computational results revealed that although the volumetric power input number of 13.5 W/m^3 is kept constant, there are quite different shear stress distributions and maximum values across scales. The difference in the hydrodynamic stress patterns was mainly due to different reactor geometries. Highest stress values are located near the impeller, and lower values are located above the impeller. In particular, for the ambr[®] 15 system, the hydrodynamic stress distributions calculated by CFD revealed that stress values are rather low in a significant fraction of the reactor volume due to the position of the stirrer very close to the bottom, which resulted in poorly mixed regions at the top of the reactor. The maximum stress values were measured or estimated (for 5,000 and 15,000 L) at each scale. It was found that the maximum values of the energy dissipation rates systemically increase with the volume, although they actually appear only in a very small region within the reactor volume. With respect to the mixing time, the obtained values increase with the reactor volume from several seconds in the 3 L system to almost two minutes in the 15,000 L bioreactor for the same specific energy dissipation rate. As observed by Villiger et al. (2018), longer mixing times are an inevitable consequence of increasing the scale. On the other hand, it is key to keep heterogeneity to a minimum in order to avoid the cells' exposure to non-physiological conditions during acidic or basic feed additions. This is even truer for continuous cultivations, as the continuous media exchange requires a full homogenisation of the cell culture broth. Accordingly, if critical and tolerated by the cells, the power input should be increased at the manufacturing scale in order to improve the reactor homogeneity.

Oxygen transfer is of crucial importance for a healthy culture. This has been investigated by combining a CFD simulation of the cell suspension with a population balance equation describing the behaviour of gas bubbles of different sizes and eventually computing the average value of the k_La . Also, in this case, quite heterogeneous patterns and values have been found between different scales. The results highlight the expected strong impact of the bioreactor configuration in terms of sparger, baffles, probes and agitation on the k_La . The study of Villiger et al. (2018) shows that modelling can be very efficient to identify operating conditions, primarily the energy dissipation rate, that optimise the reactor behaviour, for example, by mitigating inhomogeneity and gas mass transfer issues in the large scale. However, in order to be effective, this approach requires a clear knowledge of the shear stress threshold of the cell line, as discussed earlier (Neunstoecklin et al. 2015 and 2016).

Bertrand et al. (2018) performed a proteomic analysis in both 15 mL micro-bioreactors and a 300 L pilot-scale bioreactor for an industrial fed-batch platform to investigate the impact of the reactor scale on process performance, with particular reference to the intracellular processes. For this, a 17-day fed-batch CHO culture was considered. Consistency between the different scales was ensured by using similar inoculum cell densities of 0.2×10^6 cells/mL, equal standard cultivation conditions (36.5°C, pH 7.15, DO of 40 per cent air saturation) and similar feeding strategies. The time evolution of the proteome was measured, and nearly absolute accordance between the scales was verified by data-mining methods, such as hierarchical clustering, and in-detail analysis on a single protein base. While similar cell density profiles, growth rates and specific production and consumption rates were measured as a function of time independent of the reactor scale, small differences were observed with respect to the principal enzymes related to N-glycosylation resulting in slightly different glycan patterns. This could be a result of the differences in geometry, shear rate, mixing and oxygen mass transfer patterns between the ambr[®] 15 and the 300 L pilot scale. The significant fraction of poorly mixed regions within the ambr system, observed by Villiger et al. (2018), may play a significant role here. Nevertheless, it should be mentioned that the observed differences had a minor impact on the glycan distribution, and a highly comparable process performance overall was observed in the two reactors, thus confirming the reliability of the scale-up procedures.

A rather complete characterisation of a perfusion bioreactor – in terms of hydrodynamic shear rate, k_La , and mixing time – was reported by Karst et al. (2016) in the frame of a study intended to compare ATF and TFF retention devices. The ATF pump was periodically displacing 100 mL at a given flowrate of 1–1.5 L/min. The TFF set-up was driven by a bearingless centrifugal pump attached to the hollow fibre, as already discussed in Chapter 2. The reactor system was a 2.5 L benchtop bioreactor with 13 cm diameter (T). In the case of the TFF system, the glass vessel was modified with a bottom outlet to enable the direct attachment of the bearingless pump to circulate the cell culture through the external loop. A six-blade Rushton turbine impeller with a diameter, D , of 4.5 cm was used, and a four-whole-open-pipe sparger with 1 mm openings was installed 1 cm below the impeller to supply cells with the required amount of oxygen. The working volume was set to 1.5 L. The authors evaluated mass transfer coefficients for the given set-up as a function of the specific power input and superficial gas velocity at various stirring speeds (300–500 rpm), and sparging rates (0.11 – 0.33 vvm). For the given bioreactor configurations, k_La values in the range of 5 – 40 h⁻¹ were determined, which would be sufficient to support up to 10⁸ cells/mL. The

5.5 Characterisation of the Bioreactor: Case Studies

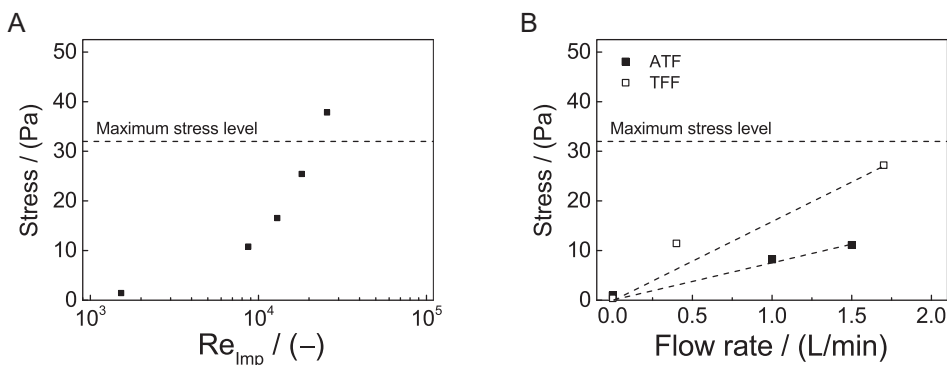


Figure 5.7 Maximum hydrodynamic stress measured in a 2 L bioreactor: (A) as a function of the impeller Reynolds number (Re_{imp}) and with no cell filtration and no sparging; (B) as a function of the recirculation flowrate in the filter in the presence of a ATF (filled squares) and a TFF (open squares) cell retention device with a sparging rate of 0.22 vvm and an agitation speed of 400 rpm. The horizontal dashed line indicates the critical shear stress value of 32 Pa for the considered CHO cell line. © Reprinted from *Biochemical Engineering Journal*, 110, Daniel J. Karst, Elisa Serra, Thomas K. Villiger, Miroslav Soos and Massimo Morbidelli, 'Characterization and comparison of ATF and TFF in stirred bioreactors for continuous mammalian cell culture processes', 2016, with permission from Elsevier

mixing times were estimated from the power number computed based on the power absorbed by the motor operating the stirrer shaft, using the empirical correlation (Equation (5.9)). The obtained values for all conditions considered were in the range of 2–6 seconds, which is sufficiently fast to avoid inhomogeneities in the system. Finally, the effective maximum hydrodynamic stress within the system was estimated using the shear sensitive PMMA particulate system illustrated above (Villiger et al. 2015). The maximum tolerable stress for the used cell line was known to be about 32 Pa. Karst et al. (2016a) evaluated the individual contribution of the filtration module on the maximum hydrodynamic stress. In the case of no filtration module and no sparging, a maximum shear stress of 37 Pa – that is larger than the one tolerated by the cells – was measured at 500 rpm, as shown in Figure 5.7A. In addition, taking the cell retention system into account, the maximum shear stress value was measured at a sparging rate of 0.22 vvm, with an agitation speed of 400 rpm as a function of the circulation flowrate in the ATF or TFF retention device, as shown in Figure 5.7B. It was found that the hydrodynamic stress to which cells were exposed during cultivation was smaller in the ATF compared to the TFF system at comparable flowrates, and the resulting maximum hydrodynamic stresses were below the reported threshold for the given cell line. Therefore, a safe culture could be expected for the entire operating range. By comparing the cell culture performance within the two set-ups, very similar culture performance and product quality were observed. However, with respect to product retention, the use of the TFF resulted in considerable protein retention compared to ATF, as discussed in Chapter 2.

From the results of this study, it clearly appears that a proper reactor characterisation is of fundamental importance for process design and scale-up, particularly with reference to the shear stress. The cell retention device brings an additional contribution to the shear stress present in the reactor system. Wang et al. (2017) found for a TFF system driven by a peristaltic pump significantly higher shear rates

5 Clinical- and Commercial-Scale Reactors

Cell specific maximum tolerable stress, τ_{max}

CFD and experimental reactor characterization

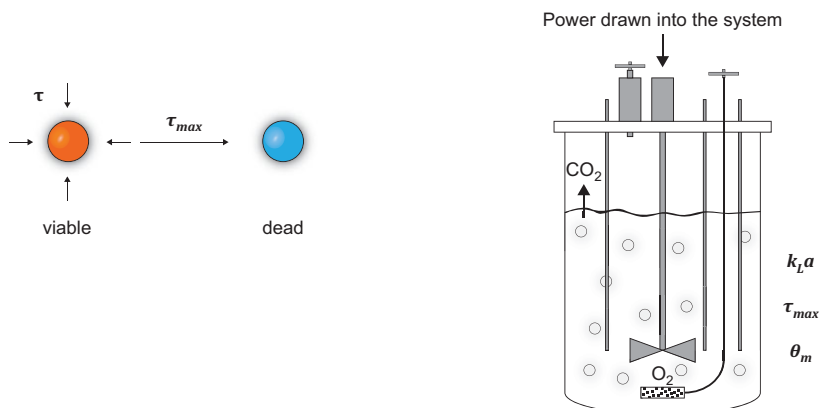


Figure 5.8 Example of the critical stress determination (dashed arrows) based on the viability drop during the first 24 h of the culture for a CHO (A) and a Sp2/0 cell line (B). Key characterisation for the reliable scale-up of a perfusion process: (1) determination of the cell specific maximum tolerable dynamic stress (Neunstoecklin et al. 2015) and (2) computational and experimental characterisation of the reactor system in terms of volumetric gas–liquid mass-transfer rate, $k_L a$, mixing characteristic time, θ_m , and maximum shear rate, τ_{max} .

compared to the ATF system or to the TFF system driven by a bearingless centrifugal pump, resulting in a higher degree of cell death and product retention. With respect to scale-up, characterisation studies for commercial-scale perfusion bioreactors are currently not available, but with the methods described in this chapter, it is definitely possible to ensure reliable scale-up to commercial scale.

The critical steps for a reliable scale-up are summarised pictorially in Figure 5.8. These include the preliminary determination of the maximum tolerable hydrodynamic stress for the given cell line and then the physical characterisation of the bioreactors at different scales in terms of interphase mass transfer, mixing and maximum shear rate.

5.6 CELL RETENTION AT LARGE SCALE

Robust and scalable cell retention devices have been a major technological difficulty that limited the large-scale application of perfusion processes for a long time. Today, commercial solutions are available at lab and large scale mainly based on tangential filtration. This is the preferred separation technology because it offers full cell retention. In addition, one of the main technical challenges is that complex mixtures such as cell cultures favour membrane fouling, and tangential flow mitigates this undesired effect. Recent technological innovations lie mostly with the filter design, either in its shape or in its operating mode. For example, filters can be designed as flat sheets or hollow fibres, to maximise the contact surface area per unit volume. The recirculation flow can be either constant and unidirectional as in TFF or it can be alternating – that is, regularly inverting its direction at a constant frequency as in ATF (Figure 5.9). Next, we address some scale-up considerations for these two filtration technologies.

5.6 Cell Retention at Large Scale

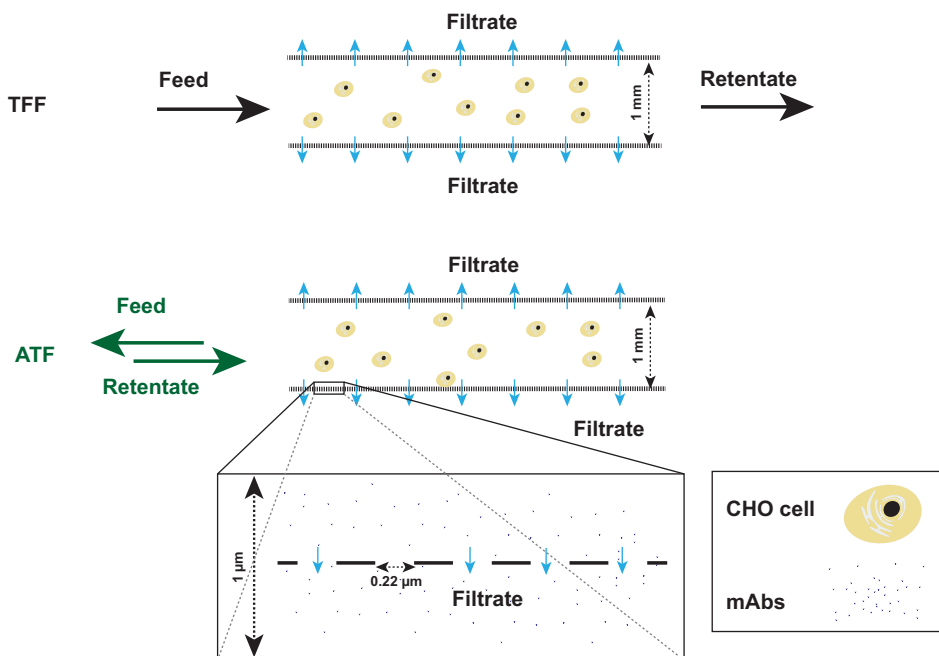


Figure 5.9 Schematic representation of hollow fibres operating in TFF and ATF filtration mode. The filter is the same, but the flow direction is either unidirectional and constant (TFF) or inverts direction at a regular frequency (ATF). The filtration direction shown by the blue arrows is perpendicular to the flow inside the fibres. The zoom on the membrane surface compares the size of the mAbs (assumed to be 10 nm large) to the pores of the membrane. The mammalian cell diameters typically range between 10 and 20 μm and are not represented at scale.

In order to scale a perfusion culture from the laboratory to the manufacturing scale, there are a few key parameters that must be carefully controlled. For example, the membrane material and porosity (usually 0.22 μm in polysulfone or polyether-sulfone) must be the same. The two main parameters to be controlled – that is, kept substantially constant – in scaling up are the residence time of the cell culture in the filtration device and the specific filtration capacity in volume of permeate per surface area of membrane and per unit time. The first one is due to the fact that the environment in the filtration device is not controlled, and therefore, the oxygen and nutrient concentrations, pH and even temperature may change with no possibility for any control action to intervene or to correct such change. The danger is that some conditions are created that affect cell viability, like complete depletion of dissolved oxygen, and this requires the cell residence time to be controlled in the filtration device. The second parameter is related to the limited filtration capacity of the membrane, which is particularly challenged at high cell densities like in perfusion cultures. The additional shear stress experienced by the cells in the filtration loop should also be taken into consideration, as discussed in the context of Figure 5.7.

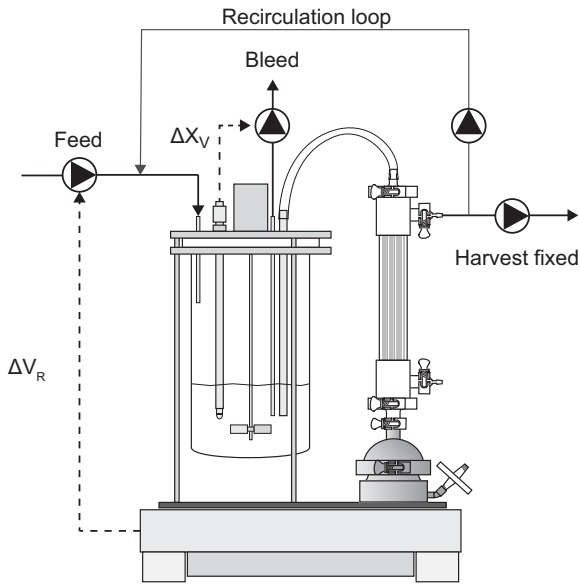
Commercial ATF systems are available for different bioreactor scales ranging from a few litres up to 1,000 L at a perfusion rate in the order of 1 RV/day. The perfusion volumes can be increased by adding filtration units and running them in parallel. The design of the filtration units for each scale, including the pump head, is

5 Clinical- and Commercial-Scale Reactors

Table 5.1 Comparison of a 2 L bioreactor with perfusion rate of 1 RV/day and equipped with an ATF2 unit (filtration surface area 0.13 m²) and a 1,000 L bioreactor with ATF10 (11 m²) and various perfusion rates. The third column reports the flowrates of the recirculation loop shown in Figure 5.10 needed to have the same filtration capacity.

Filtration capacity / L/m ² /day	ATF2 permeate flowrate for a 2 L bioreactor / L/day	Recirculation loop flowrate in Figure 5.10 / L/day	ATF10 permeate flowrate for a 1,000 L bioreactor / L/day
15.4	2	0	170
45.4	5.9	3.9	500
90.8	11.8	9.8	1,000

Figure 5.10 Perfusion bioreactor set-up with a recirculation loop to artificially increase the permeate flux through the membrane without affecting the perfusion rate. An example of how this flux can be used is described in Table 5.1.
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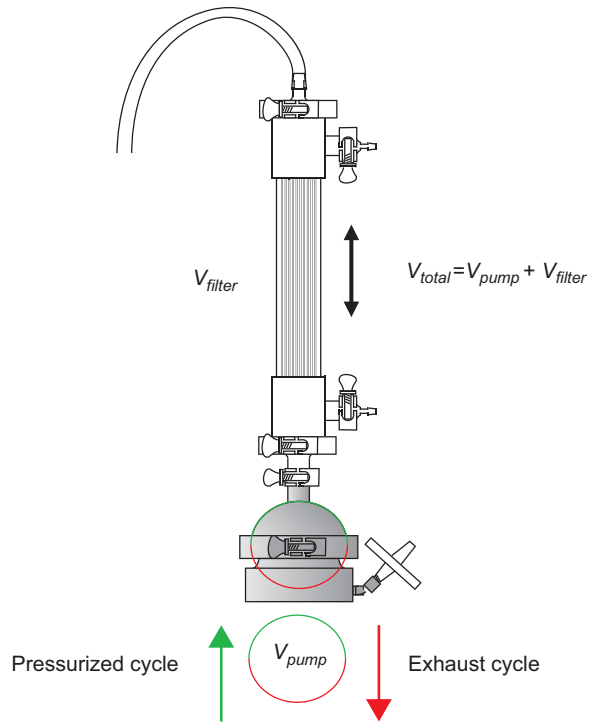
standardised and guidelines for the scale-up are usually suggested by the manufacturers. To implement ATF at commercial scale, the following parameters should be controlled:

1. Filtration capacity

The filtration capacity is given by the ratio of the permeate (harvest) volumetric flowrate and the membrane total surface area per unit of time. Because of the standardised design of the ATF, this is usually significantly oversized at the lab scale, since the total permeate flow per surface membrane is typically very low. One option to test the filtration capacity at low scale but in conditions representative of the manufacturing ones is to use a recirculation loop to pump the excess of harvest volume back into the bioreactor without affecting the perfusion rate, as shown in Figure 5.10. The application of this concept is illustrated in Table 5.1 with reference to a 2 L bioreactor equipped with an ATF2 device exhibiting a filtration surface area equal to 0.13 m² and operated at a perfusion rate of 1 RV/day. If we want to reproduce the same filtration capacity values listed in the first column of

5.6 Cell Retention at Large Scale

Figure 5.11 Schematic of a lab-scale ATF device showing the position of the membrane during the pressurisation cycle, when the liquid is pushed back into the bioreactor, and the exhaust cycle, when the liquid is sucked into the filtration unit. For the residence time calculation, the ratio of the volume of the pump and the total volume is considered, as described in Equation (5.15).



the table, as in a 1,000 L bioreactor equipped with an ATF10 device exhibiting a filtration surface area of 11 m², operated at different perfusion rates (last column in the table), we need to apply the permeate and external loop flowrates listed in the second and third columns of the table, respectively. When testing this parameter before scale-up, the same duration and total amount of permeate per surface area as in the large scale should be considered.

2. Residence time

The average residence time in a vessel of any geometry operated at steady state can be calculated as the ratio between the vessel volume and the inlet (or outlet) volumetric flowrate. For the ATF, this depends on the alternating cycle frequency and on the volume exchange ratio, defined here as the ratio between the displaced volume inside the device, equal to the pump head volume (Figure 5.11), and the total ATF volume. This ratio is typically close to one because the pump head volume is significantly higher than the volume of the hollow fibres and the connection tubing (that are never empty) and therefore accounts for most of the total volume. Accordingly, for an ATF unit, the average residence time, which is intended to be kept constant during scale-up across scales, is given by

$$\tau = \frac{V}{Q} = \frac{V_{total}}{V_{pump} \times f}. \quad (5.15)$$

3. Shear stress

Shear stress mainly depends on the alternating flow velocity inside the fibres, which should be kept constant across scales. This comes, of course, in addition to the pump specific shear rate.

4. Backflush

During the operation of an ATF unit, it is possible that the permeate flux changes direction; this is referred to as backflush, where some permeate goes back inside the hollow fibres. This has a beneficial effect on the filter operation since it helps to maintain the cleanliness of the membrane by reducing fouling. The permeate flux direction and flowrate depend on the transmembrane pressure (TMP) – that is, the pressure difference between the retentate and the permeate side of the hollow-fibre membrane. In normal operation, the pressure gradient decreases towards the permeate side, but when this is reversed, then the flux direction changes, leading to the backflush. In ATF, because of the alternating operating mode, the retentate flux continuously changes direction and velocity during each cycle, and therefore, the TMP changes accordingly along the hollow fibre and in time. With a higher circulation rate in the ATF, the maximum tangential velocity is larger, and the TMP gradients are also larger, resulting in a stronger backflush. Because of the design of the ATF, the backflush is constant if the ratio between the permeate rate and the ATF rate remains constant. Therefore, the external recirculation loop shown in Figure 5.10 can also be used to change the ratio between the ATF rate and the permeate flowrate to reproduce the manufacturing conditions.

5. Filtration device-to-reactor volume ratio

This ratio is proportional to the average number of times that a cell travels through the retention device, and it should not change significantly across scales. This is probably the most difficult parameter to mimic at small scale, where it tends to be very large. Since this is the worst case with respect to cell behaviour, the manufacturing scale is usually not impacted by this parameter. In general, the most advisable strategy for the reliable scale-up of an ATF device is to keep constant the length of the hollow fibres while properly increasing their number. This enables constant pressure drop across the filtration package, which in turn implies constant liquid flowrate, constant average residence time and constant shear stress. On the other hand, increasing the fibre length (with constant number) would lead to higher back pressures and consequently larger shear stresses. This effect was demonstrated at lab-scale by Karst et al. (2016).

Commercial TFF systems, equipped with some devices for the control of the recirculation and the harvest streams, are available on the market for lab-scale bioreactors. At the manufacturing scale, filters like the ProstagTM flat sheet from Millipore offer the possibility to have enough filtration surface, but the recirculation system is not provided. An external skid, including all the necessary pumps and control systems to monitor and control the recirculation loop, must be designed ad hoc. To implement TFF at manufacturing scale, the following parameters should be controlled:

1. Filtration capacity

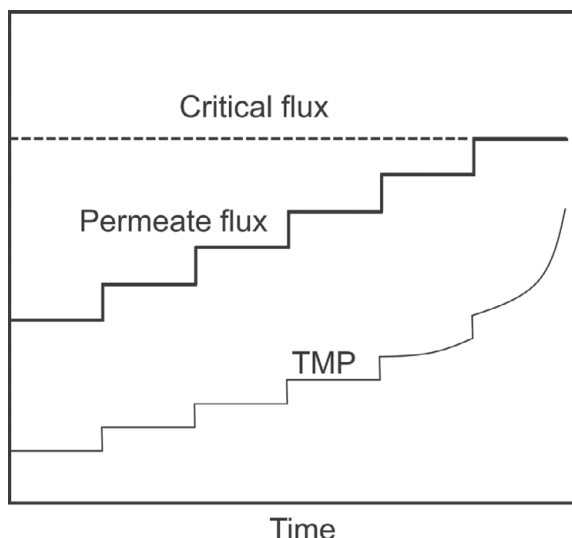
This parameter should be tested at the small scale, before scale-up, for the same duration considered for the bioreactor design and for the same amount of permeate per surface area.

2. Residence time

The average residence time can easily be computed as the ratio between the total free volume of the filtration loop and the circulating volumetric flowrate (Figure 5.13).

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Figure 5.12 Experimental procedure to define the critical flux. The permeate flux is increased stepwise, and the corresponding TMP increase is monitored. The critical flux is reached when the TMP is not stable anymore and increases strongly due to irreversible fouling.



3. Shear stress

Shear stress should be minimised by avoiding all possible sources of pressure drop in the circulation loop, like valves, changes in cross section, pipe turns or others, so as to minimise the energy that the pump has to provide to the suspension. In this regard, the type of pump, its geometry and the shear stress imposed to the suspension need to be controlled – that is, remained unchanged upon the change in scale.

4. Critical flux

This is an empirical concept described by Bacchin et al. (2006) indicating the critical permeate flow, above which irreversible fouling occurs and the device cannot be operated any longer. The critical flux corresponds to the transmembrane pressure at which the system becomes unstable due to fouling, as described in detail by Raghunath et al. (2013). Any flux value below this critical value will not impact the TMP, and fouling is reversible. In tangential flow filtration, the critical flux is influenced by the tangential flux velocity, which depends on the stepwise increase and the corresponding value of the TMP is measured as shown in Figure 5.12. The flux value at which the TMP starts to increase sharply over time reflects the appearance of fouling and is taken as the critical flux recirculation rate (Bacchin et al. 2006, Beier & Jonsson 2009). This parameter can become limiting in high-perfusion-rate operations, such as those typically used in N-1 applications.

5. Filtration loop to reactor volume ratio

This ratio represents the number of times the cells visit the filtration device and the same considerations reported earlier for the ATF unit hold true.

Considering that there is no standard equipment available on the market, the implementation of TFF units probably needs extra precautions to ensure that all the parameters described earlier are within a feasible operating range. Special care must be given to the configuration of the recirculation loop to avoid undesired shear stress contributions or pressure gradients that could impact the performance of the entire perfusion system. For the same reasons, these systems provide more flexibility for the design of specific processes compared to the ATF counterpart.

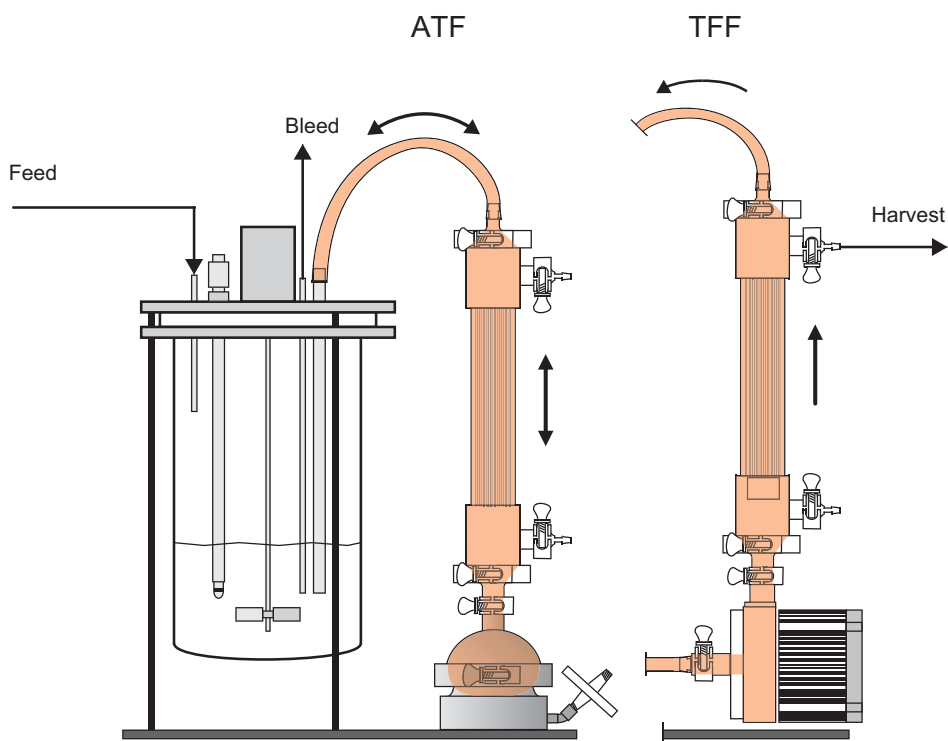


Figure 5.13 Schematic representation of laboratory ATF and TFF filtration modules: the orange color indicates the volume to be considered in the calculation of the average residence time, which is changing in the case of ATF, as described in Figure 5.10. For the TFF this includes all tubing from and to the bioreactor.

Different kinds of failures may occur when operating filtration cell retention devices, mainly due to the complexity of the suspension to be filtrated, which in perfusion bioreactors is particularly concentrated and viscous. Membrane fouling and clogging are the primary concerns, with the suspension containing small and large molecules, hydrophobic and hydrophilic compounds, cells and cell debris, and so on. On the other hand, it is also possible that due to some transport difficulties, particularly in the presence of some partial fouling, some of the target protein is also retained inside the reactor. This undesired event is referred to as sieving in the literature (Lin et al. 2017, Radoniqi et al. 2018, Wang et al. 2017) and is quantified as follows:

$$\text{Sieving} = \left(1 - \frac{C_{\text{Harvest}}}{C_{\text{Reactor}}} \right) \times 100, \quad (5.16)$$

where C_{Harvest} represents the protein concentration in the harvest and C_{Reactor} represents the protein concentration inside the bioreactor. If the two are equal, then no target protein retention occurs, and the sieving effect is zero. Fouling and sieving may represent serious problems in scale-up, forcing the limitation of the operation duration or the cell density. A quantitative understanding of these processes is not available, and experimental reports are not always coherent. In general, it is believed that the alternating flow typical of ATF creates a backflush on the filtration membrane

5.7 Process Intensification

that helps to prevent fouling over longer periods of time (Bonham-Carter & Shevitz 2011, Kelly et al. 2014, Radoniqi et al. 2018). Karst et al. (2016) characterised and compared two TFF and ATF units under similar conditions using a bearingless centrifugal pump for the TFF and reported up to 50 per cent product retention when using the TFF and none with ATF. Clincke et al. (2013b) tested the limits of these two systems in wavebag cultivation systems. It appeared that the highest cell density obtained was around 210×10^6 cells/mL with the TFF system, although this density could not be sustained for longer times. Using the ATF, the maximum reached cell density was only 130×10^6 cells/mL because of the high viscosity that was limiting the liquid displacement capacity of the ATF pump. The two studies mentioned here show that both systems have different advantages and limitations that are yet to be fully understood.

In the case where fouling or sieving cannot be prevented for a given process during longer operations, it is possible to connect several filtration units to either use them in parallel or as a backup if necessary. These strategies are indeed viable, but care must be taken not to increase the contamination risk. An interesting alternative is offered by pre-sterilised and single-use devices that are described in Section 5.8.

5.7 PROCESS INTENSIFICATION

Pharmaceutical industry is experiencing unprecedented pressure from the market because of the increasing pressure on cost, the uncertainty and variability of the product demand, the market growth and regionalisation and the emergence of new classes of products. To face these challenges, alternative technologies are needed to improve the quality of the product, reduce the costs, increase the manufacturing flexibility, increase the speed to market and decrease the release time. These are crucial to answer the demand for modern biotherapeutics and prevent drug shortage or other manufacturing or supply limitations. Among many enabling technologies, some were identified by the BioPhorum Operation Group (BPOG), an international group of industrials that is collaborating to guide the technological changes in biomanufacturing. These include process technologies, automated facilities, modular and mobile facilities, in-line monitoring and real-time release, knowledge management and supply partnership management (Sawyer et al. 2017b).

In the following, we focus on process intensification, which is largely driven by continuous manufacturing solutions like perfusion. Nevertheless, perfusion alone is not an answer to the previously mentioned challenges and must be seen in a larger context. As an example, in-line monitoring techniques, discussed in Chapter 3, are crucial for controlling continuous processes but could also enable real-time batch-release procedures. These would, from a regulatory perspective, ensure the quality of the final drug substance or drug product and be valid to release a production batch. This brings along a significant number of advantages, including better process understanding, error reduction and improved product consistency by closed-loop feedback control systems. Finally, the lag time between production and actual release could be significantly reduced from weeks to days. Long lead times due to analytical testing before the release are indeed a major bottleneck (Jiang et al. 2017, Swann et al. 2017).

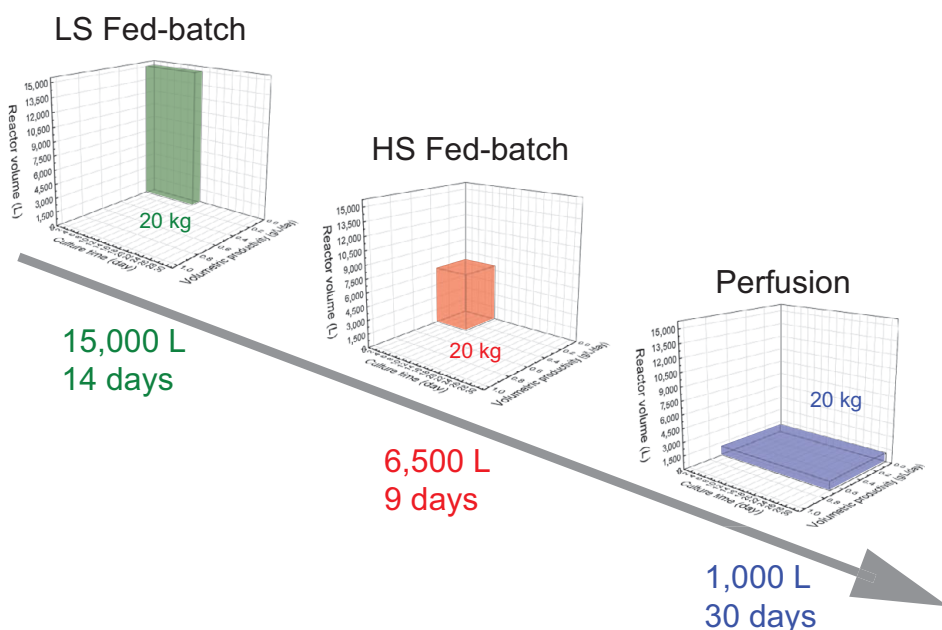


Figure 5.14 Footprint of different process types for the production of the same amount of target protein, based on the case study presented in Chapter 4, Figure 4.19. The production volumes are 15,000 L, 6,500 L and 1,000 L for traditional fed-batch, intensified fed-batch and perfusion, respectively. A relevant parameter is the run duration which is longer for the perfusion process.

To intensify existing fed-batch processes, the seed train can be intensified by introducing a perfusion bioreactor before the production bioreactor (N-1) to increase the inoculated biomass. This is done in the so-called high seeding fed-batch processes described in the example in Section 4.5.3. The objective is to increase the volumetric productivity in the production bioreactor (Bausch et al. 2018, Chen et al. 2018), which in general results from the higher seeding density leading to shorter runs with higher final titres. On the other hand, operating the production bioreactor directly in the perfusion mode can further increase the volumetric productivity, since high cell density values are maintained during the entire operation for longer times. This is well illustrated by the example of the conjugated protein described in Section 4.5.3, where it is seen that, for a fixed total production of the target protein, the bioreactor volume could be decreased from 15,000 to 6,500 L using batch-like intensified processes and even further to 1,000 L operating a perfusion bioreactor at steady state for 30 days, as shown in Figure 5.14. The comparison of the three operation modes was performed under platform conditions, which means that under optimised conditions, the specific productivity could be improved even further. Chen et al. (2018) described a hypothetical example using reasonable assumptions leading to a footprint reduction from 15,000 to 2,000 L for intensified fed-batch and to even 500 L for perfusion.

In the following, the implementation of perfusion for intensification of existing processes and end-to-end continuous integrated manufacturing is discussed through a few relevant examples. The value of single-use technologies that are often combined with these new manufacturing strategies is discussed in the next section.

5.7 Process Intensification

Table 5.2 Freezing container type and corresponding maximum viable cell density values. © Reprinted from *Biotechnology Advances*, 36/4, Jean-Marc Bielser, Moritz Wolf, Jonathan Souquet, Hervé Broly and Massimo Morbidelli, ‘Perfusion mammalian cell culture for recombinant protein manufacturing: A critical review’, 2018, with permission from Elsevier.

Reference	Freezing container	Maximum Cell density / (10^6 cells/mL)
Heidemann et al. (2002)	50 or 100 mL cryobags	20 or 40 respectively
Tao et al. (2011)	5 mL cryotubes	90–100
Seth et al. (2013)	150 mL cryobags	70
Wright et al. (2015)	4.5 mL cryotubes	100

5.7.1 Process Intensification Using Perfusion

There are multiple ways to integrate perfusion to support process intensification in general. As discussed by Bielser et al. (2018), these range from high-density cell banks to the optimisation of the N-1 bioreactor in front of the production bioreactor and, finally, to the actual replacement of the fed-batch production bioreactor with a continuous perfusion one.

Table 5.2 summarises some literature references describing cell banks for high cell density inoculation. Here, perfusion technology is used as a tool to concentrate a culture with the cells in the exponential growth state. They are then frozen in either standard cryotubes (a few mL) or larger cryobags (50–150 mL). Using these concentrates, the seed train prior to a production bioreactor can be significantly shortened. From a manufacturing perspective, this is important not only because of the production time reduction, per se, but also because this allows a higher degree of flexibility at the level of production planning. It is also worth noting that, using cryobags, all operations can be performed in closed systems (no need for a laminar flow hood), which reduces significantly any contamination risk.

As already mentioned, another application of perfusion is at the level of the N-1 bioreactor located in front of the fed-batch production unit (Chen et al. 2018, Jordan et al. 2018, Pohlscheidt et al. 2013, Yang et al. 2014). In this case, medium is perfused to increase the cell density as much as possible while keeping the cells in an exponential growth phase before transferring them to the production bioreactor. The bleed outlet stream is not necessary in this case since the goal is not to reach steady-state conditions but simply to maximise viable cell density to seed at higher concentrations than in traditional fed-batch operation. This leads to shorter production runs and, consequently, lower occupancy of the production bioreactor. This again offers significant advantages in terms of planning, flexibility and, therefore, productivity, with the additional advantage that this technology can be implemented in existing facilities, originally designed for batch operation, including production bioreactors larger than 10,000 L.

Perfusion can also be used only in the initial phase of a fed-batch production. That is referred to as a hybrid process. Hiller et al. (2017), for example, describe a process in which perfusion was applied during the first 4 days only. The maximum viable cell density was increased to 60 or 80×10^6 cells/mL and the productivity increased up to 2.5-fold, depending on the clone, with respect to classical fed-batch operation. This example shows that the combination of batch-type processes with perfusion can often lead to improved performance. Perfusion could, for example, also

be applied at the end of a fed-batch process to remove excessive toxic waste products, maintain a higher viability for a longer period, and even clarify the culture and eliminate standard centrifugation and depth filtration steps (Bonham-Carter 2018).

The forementioned examples show only a few of the options that one can think of to couple continuous and traditional fed-batch operation to obtain intensified processes with superior performance. Whenever possible, these are valuable options and should be pursued. However, the implementation feasibility will depend on the constraints resulting from the existing plant. To upgrade a production or N-1 bioreactor to be operated under perfusion conditions, a cell retention device and the underlying pumps for recirculation must be added. This means that enough space at the right location must be available. The media production capacity must also either already exist or else be created for that purpose. Finally, automation to control the perfusion rate is also required. The upgrade of an existing production suite is a significant investment, but considering the potential advantages demonstrated extensively in the literature, it can be an interesting option to consider.

Using a single perfusion bioreactor at the production stage is the ultimate implementation step for this technology. The complexity of the system is increased because it requires an additional control on the bleed stream to achieve steady state. The medium volumes are, compared to N-1 runs, much more important, and comparable harvest volumes need to be treated by the downstream purification units. In order to avoid storing huge volumes of harvest, the best option is therefore to operate also the capture and purification units in continuous mode. This type of operation is described hereafter.

5.7.2 End-to-End Continuous Integrated Operation

Continuous integrated protein manufacturing refers to a process where the upstream continuous bioreactor is coupled with a continuous downstream purification section and the two are operated as a single unit (Karst et al. 2018). In the classical case of monoclonal antibody production, this translates into a rather general production platform, where the perfusion bioreactor is followed by protein A capture and a couple of polishing steps, in addition to viral inactivation, as illustrated in Figure 5.15 (Baur et al. 2016, Pfister et al. 2018, Steinebach et al. 2016b, Vogg et al. 2018). The first successful demonstration of the integration of a perfusion bioreactor and a continuous capture step was reported by Warikoo et al. (2012) for

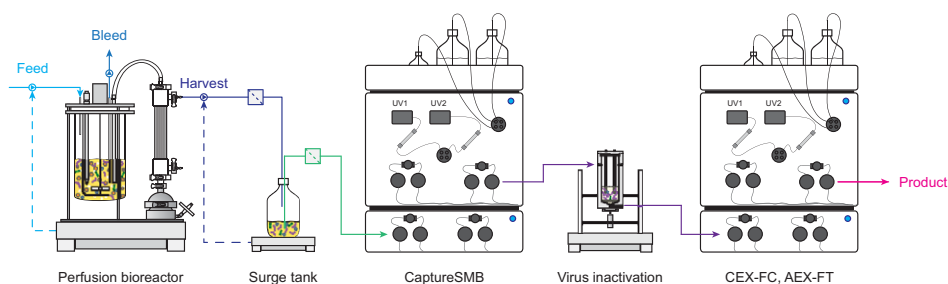


Figure 5.15 Continuous end-to-end process schematic, typical for monoclonal antibody production, including a perfusion bioreactor connected to a capture two-column simulated moving bed unit and then to viral inactivation and two polishing steps.

5.7 Process Intensification

the production of a mAb and a recombinant human enzyme, which correspond to a stable and a complex and unstable protein, respectively. Besides any other engineering consideration, this demonstrated for the first time the dramatic reduction of the production unit footprint because of the elimination of intermediate hold tanks. This result probably marked the very beginning of continuous manufacturing in biopharmaceutical industries, which most scientists in this area now agree to have very attractive perspectives.

One crucial turning point in the development of this technology will be the massive introduction of automation and digitalisation, based on local sensors connected to supervisory control and data acquisition (SCADA) units which, properly supplemented with mathematical models, can run the unit, rejecting disturbances and keeping optimal performance, as discussed in Section 6.7. Each unit operation can be operated at its best condition, while assuring proper communication between the different units. This would have significant impact not only on the overall production costs but even more importantly on the product quality and safety, with eventual benefits to the patients.

A few demonstrations of end-to-end integrated processes, which also included viral inactivation and polishing steps, have been reported in the literature. Godawat et al. (2015) described an end-to-end process that included perfusion, continuous capture and post-capture downstream processing. They demonstrated over a longer period of time (30 days) that such a process could deliver consistent productivity and product quality. The process included a perfusion bioreactor and two periodic counter-current chromatography systems for capture and polishing, with a viral inactivation hold step in between. In terms of performance, the volumetric productivity was increased tenfold and sixfold, compared to batch operation, for upstream ($\text{g/L}_{\text{Reactor}}/\text{day}$) and downstream ($\text{g/L}_{\text{Resin}}/\text{day}$), respectively. The column size of the chromatographic steps was reduced 20-fold, and the buffer consumption reduced by 20 per cent.

A lab-scale integrated process for the production of a mAb was reported by Steinebach et al. (2017). The process included a perfusion bioreactor, a two-column continuous capture step, a two-column continuous (bind and elute) polishing step, viral inactivation and a final batch flow-through polishing step. A constant product output – in terms of concentration, aggregate and fragment content but also charge isoform and glycan distribution – was demonstrated. The contribution of each downstream step in reducing the impurity content in terms of HCP, aggregates, fragments, DNA, and protein A leached in the capture step was discussed.

An alternative operation mode has been introduced by Coffman et al. (2017) by using a short-term (<15 days) non-steady state perfusion with low perfusion rates (0.3 RV/day) coupled with an online UPLC in order to monitor product quality and titre. The perfusion bioreactor is integrated with two batch chromatographic steps, a continuous virus inactivation step, and avoids in-process pooling. In this set-up and the product is stored after the second chromatographic step for the duration of the complete batch (<15 days) and then pooled and batched through a virus reduction filter and ultrafiltration diafiltration (UFDF) towards the bulk drug substance.

Arnold et al. (2018) demonstrated the feasibility of end-to-end continuous manufacturing at pilot scale for the production of a recombinant mAb. Compared to the reference fed-batch process, comparable levels of aggregates, host cell proteins and DNA content could be achieved. They used the BioSolve software to compare

the productivity of both processes and demonstrated that a fed-batch facility with $4 \times 12,500$ L stainless-steel bioreactors could be replaced by $5 \times 2,000$ L single-use perfusion bioreactors, each including its respective batch and continuous purification train, with an operating cost of goods reduction of 15 per cent.

The implementation of quality control as indicated by the regulatory authorities requires the definition of a batch – that is, a finite portion of product to which certain measured quality attributes refer. In the case of continuous manufacturing, this is not an obvious concept as it is for traditional batch operations. In general, a batch in continuous processing can be defined as the amount produced in a fixed interval of time, suitably defined in agreement with the regulatory guidelines (Allison et al. 2015). Failure management can also benefit from diverse batch definitions. In the case where, for example, a batch is defined at regular intervals in a continuous process, the product is secured on a regular basis. A contamination issue that occurs in the middle of a run would, for example, not impact the fractions that were released during the preceding portion of it.

Thus summarising, end-to-end continuous manufacturing is basically combining different enabling technologies such as continuous upstream and downstream, single-use equipment, in-line monitoring, automation and supervisory systems. One obvious benefit of an end-to-end continuous process is the removal of all the hold tanks present in batch operations, thus significantly reducing the production suite footprint. The bioreactor size is decreased because of process intensification, and the continuous downstream process allows the implementation of more efficient counter-current chromatographic processes (Pfister et al. 2018). Together, these technologies can fit in a modular production suite that can be easily parallelised and operated in a so-called ballroom design, which is defined as a manufacturing area with no fixed equipment and minimal space segregation. A further benefit could be obtained by confining the different units in closed systems. In this case, the entire zone could be unclassified, and the cost of the manufacturing space would be dramatically reduced because of common operating costs such as heating, ventilation, air conditioning and, finally, as already mentioned, the room classification (Kratzer et al. 2017, Wolton & Rayner 2014). Continuous end-to-end manufacturing is a perfect candidate to implement this concept, especially combined with single-use technologies that are addressed in the next section.

5.8 SINGLE-USE TECHNOLOGY

Single-use equipment is very popular in biomanufacturing for a number of different applications. For cell culture, it has been used for many years at the laboratory scale, while larger scale equipment has been developed more recently and is rapidly gaining acceptance (Shukla & Gottschalk 2013, Shukla & Thömmes 2010). Indeed, it offers distinct advantages which are regarded as crucial for future manufacturing processes. These include lower production costs, higher flexibility, decreased footprint and its being ideal to cope with market regionalisation (Chen et al. 2018, Sawyer et al. 2017b). Single-use equipment allows the reduction of the capital cost investment for a new plant construction and commissioning. Costs related to cleaning, validation and utilities are reduced, and rapid changeover and reduction of cross-contaminations make this equipment ideal for operation in a multiproduct site (Shukla & Gottschalk 2013).

5.8 Single-Use Technology

Figure 5.16 A 2,000 L Mobius single-use bioreactor system. A stainless-steel holder is necessary to contain the pressure inside the single-use bag when filled with liquid. A central unit monitors different signals and controls agitation, aeration, temperature, pH and the different pumps. Courtesy of EMD Millipore Corporation.



A single-use bioreactor must, by definition, be made of an FDA-approved plastic that is qualified for the manufacturing of drug products (e.g., polyethylene, ethylene-vinylacetate, polycarbonate, polystyrene) and can be used only once (Löffelholz et al. 2014, Mokuolu 2018). They exist at different scales and up to 2,000 L, as the one shown in Figure 5.16. It has been shown that single-use bioreactors can perform equally well as glass or stainless-steel bioreactors (Smelko et al. 2011). Nevertheless, there are some limitations that must be overcome to ensure proper utilisation of these equipment. The most critical are probably the leachable/extractable, potentially toxic or inhibitory compounds that these materials tend to release into the cell culture. Many studies report this problem, and the impact on cell culture growth has even been demonstrated in some cases (Dorival-García & Bones 2017, Hammond et al. 2014, Kelly et al. 2016). A second important limitation factor is the supply chain. There is, in fact, only a limited number of vendors on the market, and the quality and security of the supply is a high risk if it is not under control. To mitigate these risks, equipment standardisation and well-established relationships with vendors play an important role (Barbaroux et al. 2014, Shukla & Gottschalk 2013). On the other hand, it is also expected that if the global tendency of the industry is to move towards single-use equipment, the vendors will be in a more comfortable position to secure the supply in a more stable and sustainable manner (Kolwyck et al. 2017).

Other limitations for single-use bioreactors include the scale (currently limited to about 2,000 L), and the high cost of disposable items and waste disposal. Nevertheless, the scale limitation is in line with the plant footprint reduction and the resulting increase of flexibility, which constitutes one of the main advantages of this technology.



Figure 5.17 ATF units at different scales: the three units on the right are stainless steel and must be autoclaved before use, while the three on the left are single-use cell retention devices that include the pump head. XCell™ ATF Cell Retention Devices from Repligen. Photo provided by Repligen Corporation, all rights reserved

We have already mentioned that TFF and ATF filtration are available on the market as single-use devices for large-scale manufacturing. They offer similar advantages as single-use bioreactors in the sense that they are pre-sterilised and ready to use. Of course, the filter itself is anyway single-use, but in this case, the entire set-up, including pump head and tubing, is single-use. In the case of the non-single-use ATF system, the filtering membrane is first assembled with the pump head and the connection lines, and then the whole set-up is autoclaved together, which may be cumbersome due to the size and weight of these units. Having access to a ready-to-use system with the pump head, filter and all connection lines pre-sterilised reduces contamination risks and operational constraints, in addition to eliminating the need for a large autoclave. Both devices the stainless-steel and single-use ATF, are illustrated in Figure 5.17.

Besides bioreactors and cell retention devices, a wide range of single-use equipment exist and include media storage and preparation bags, sterile connectors, single-use probes, filters, pre-packed columns for downstream application and many others. In principle, each of these can be used to realise a fully single-use end-to-end integrated process.

5.9 ECONOMICAL DRIVERS

It has been mentioned several times that continuous bioreactors may have a beneficial effect over investment and operation costs compared to fed-batch units in different manufacturing scenarios. Indeed, this issue has been treated and analysed in the literature, with a number of valuable contributions coming from the industry. Particularly relevant here are the efforts of the already mentioned BPOG, who has generated a number of reports that discuss different scenarios of the future of biomanufacturing (available online at www.biophorum.com). The first conclusion is that there will be several types of manufacturing facilities depending mostly, but not entirely, on the product demand. Large-scale stainless-steel facilities for fed-batch will probably

5.9 Economical Drivers

continue to be competitive for very large production volumes. Perfusion in single use is rather expected to be favoured in the case of medium throughput production of different molecules. This corresponds to a facility that can be easily reconfigured and to suites scaled across or, in other words, copied and parallelised (Sawyer et al. 2017b).

Of course, as mentioned before, the comparison between fed-batch and perfusion technologies depends on the specific industrial application under examination and particularly on the requested production capacity. However, the latter is not always well defined for biopharmaceuticals, or at least much less than for other products on the market. The development time lines of a successful drug can, in fact, take more than a decade, and the failure risk is extremely high. The clinical outcome during the clinical phases and even during the product lifetime play a significant role on the demand uncertainty.

As an example, immuno-oncology is a characteristic treatment field of biopharmaceuticals in which molecules are used to activate the immune system against a specific cancer (Hoos 2016). These drugs are highly promising not only because of their efficiency but also because they target only the cancer cells and do not harm the entire body as is typically the case for small molecule chemotherapies (Antonia et al. 2014, Chen 2013, Fisher et al. 2018, Kohrt et al. 2016, Lameris et al. 2014, Solomon & Garrido-Laguna 2018). Therefore, the demand for these drugs increases continuously, and their prices, which are also related to political decisions, are continuously under pressure (Howard et al. 2015, Sawyer et al. 2017b). All of this provides the motivation to provide ever-increasing amounts of these drugs to people at lower prices but without compromising efficacy and safety.

A variety of models were proposed to understand the economic impact of perfusion and, more generally, continuous processing. Pollock et al. (2013) developed a model to compare traditional fed-batch with perfusion in different industrial scenarios. In this case study, not only the production bioreactor but also the related purification steps were considered. Two perfusion processes were imagined: a first- and a second-generation one using a spin filter and an ATF as cell retention devices, respectively, with the second allowing for much higher cell density values. Of course, this kind of modelling requires many assumptions, but the conclusion was that using an ATF perfusion strategy, and assuming a fivefold increase in maximum cell density compared to fed-batch, the cost of goods (COG) savings could be up to 20 per cent. The model includes investment and consumable costs, and the COG calculations were made for different yearly productions (e.g. 100, 500 and 1,000 kg/year). With this model, considering only costs of goods, the ATF scenario is always beneficial in terms of cost. In the same study, the structure of the COG was also investigated. Without any surprise, the cost of bags and media came as the major contributors to the overall production cost.

Walther et al. (2015) developed a complex model for integrated continuous biomanufacturing with a dynamic scenario in which multiple products (stable and unstable proteins) with different annual demands are launched over 10 years. The model uses processes-economic modelling tools, Monte Carlo simulations and risk-based net present value (NPV) analysis. The overall learning for the investigated cases is that integrated continuous manufacturing offers significant financial advantages, in terms of both capital and operational costs.

Another study came to a different conclusion. Klutz et al. (2016) estimated the COGs for fed-batch and perfusion, including batch and continuous downstream,

respectively. The result was that for the upstream part, the cost of goods of perfusion was systematically higher than for fed-batch. For downstream instead, continuous operation was always cheaper than batch-mode purification. Therefore, Klutz et al. (2016) suggested a hybrid process with fed-batch cell culture followed by continuous purification. This is indeed a viable strategy that can be particularly interesting in the case of existing facilities that encounter capacity limitations on the downstream side.

A relevant aspect when considering production costs refers to cell culture media, which accounts for a significant portion of the cost of goods and usually offer quite some room for optimisation (Langer 2011, Xu et al. 2017b). Media management for perfusion is not a trivial question and has already been addressed in Section 2.5. In principle, each manufacturer can choose between commercial and proprietary media. If commercial media are used, the value of the minimum cell-specific perfusion rate ($\text{CSPR}_{\min}$) for the used cell line is fixed, and this defines the amount and composition of media needed with no room for further optimisation. With proprietary media, each compound can potentially be balanced to the minimum required level for a targeted CSPR. Ideally, in fact, all the compounds fed to the bioreactor should be consumed and transformed in either energy, biomass or target product. The harvest would, in this case, only contain water, metabolic waste products and the produced protein. It is obviously difficult to adjust all media components so precisely, especially as some of them like salts and pluronic are not meant to be consumed. Nevertheless, there is significant room for the cost optimisation of media composition, particularly when considering that the different components have quite different costs. Accordingly, in media development, one should always consider the price of each compound and contrast this with the benefit it brings to the culture. This work must be done early in the process development since it can lead to significant financial benefits.

In industrial practice, it should be considered that the cost of media is not only driven by the raw material value but also by the cost of preparation and storage. Consider the case where a 1,000 L bioreactor is operated at one reactor volume per day for 30 days. This means that 30,000 L of media must be prepared during such 30 days, and the question arises about what the most economic strategy is. If the entire batch of 30,000 L is prepared in advance, then huge storage tanks are necessary, which is obviously not in line with the plant footprint reduction that is envisaged for future technologies. Alternatively, one could prepare a defined volume every day or second day, but this increases operational burden and still requires some storage capacities. Other scenarios include online dilution of a concentrated feed with some buffers and online powder dilution. This is clearly not a trivial problem and further justifies the efforts that were discussed in Section 2.5 to decrease as much as possible the media consumption, which means the $\text{CSPR}_{\min}$ value (McCoy et al. 2015, Xu et al. 2017b).

Besides economics, it is also important to evaluate the impact that different manufacturing technologies have on the environment. Bunnak et al. (2016) performed a life-cycle assessment (LCA) of the bioreactor operating mode by quantifying its environmental impact in terms of water consumption, energy requirements and solid waste generation. It is found that perfusion has a larger environmental impact than fed-batch, with respect to all parameters considered: 35 per cent more water, 17 per cent more energy and 17 per cent more CO_2 emission. However, the outcome of this study depends significantly on the selected downstream strategy. The perfusion

5.10 Conclusion

harvest in this study was assumed to be pooled for four days before being further processed in batch-mode downstream purification. When this pool duration was increased to eight days, the two bioreactor operating modes were found to exhibit the same water and energy consumption. These results indicate how the environmental footprint can be sensitive to different choices about manufacturing technologies.

In conclusion, it is worth mentioning that a better understanding of the economic, financial and environmental aspects of biomanufacturing is quite relevant to guide the development of innovative technologies including those based on continuous operation. This is an essential step towards new manufacturing strategies that can bring to the market significantly safer and cheaper therapeutic proteins.

5.10 CONCLUSION

In this chapter different aspects of the scale-up of perfusion cell cultures from the laboratory to the clinical and commercial scales are discussed. The three most relevant scale-up parameters related to aeration, mixing and shear stress are discussed in detail. Scale-up characterisation and validation is then analysed with close reference to both the essential components of a perfusion system: the stirred bioreactor and the cell retention device.

Next, we analyse the role that perfusion can play in the intensification of large-scale production facilities. This is seen in the context of existing facilities based on batch-type technologies as well as with reference to integrated continuous end-to-end units, which indeed provide a fascinating alternative and a paradigm change with respect to the current biomanufacturing technologies.

A lot of innovation is currently ongoing in biomanufacturing, and most of it is ready or getting ready to be transferred to the clinical and commercial scales. This includes continuous integrated manufacturing, single-use equipment, online sensors, supervisory control and digitalisation. All of these could be combined in a single, end-to-end integrated and highly automated manufacturing platform (Fisher et al. 2018). This would probably include perfusion bioreactors between 200 and 1,000 L, which could be parallelised if needed, with high flexibility for demand fluctuations or multiproduct productions, always keeping a high level of quality control. Closed operation would minimise contamination risks and could even lead to declassified manufacturing areas, with substantially lower investment costs. By decreasing the footprint, cost and complexity of the biomanufacturing units, market regionalisation could be addressed more easily by spreading the manufacturing capacities across the world, and avoid centralised production strategies, as it often happens today (Moyle 2017).